Quantitative Hamiltonian path counts and Wormald's angular Hessian

Xingchen Liu and Xiangyu Ye

7 November 2026

Abstract. We prove that the maximum number of directed Hamiltonian paths in an n-vertex tournament satisfies

$$P(n) = \left(\frac{\cosh 1}{\cos 1} + O(n^{-1}) \right) \frac{n!}{2^{n-1}}.$$

The proof builds on our earlier constant-factor counting framework, combining uniform permanent estimates, local diagonal scaling for nonprincipal submatrices, and a score-sensitive path bound with the spectral comparison of Fang and Huang. Divoux and Wang established the sharp leading asymptotic constant for cycles and paths. We also give an independent derivation of the full angular Hessian in Wormald's random blow-up formula and prove its positive definiteness for every regular base. The overlap with Bhattacharjee's results is stated explicitly. For each fixed odd base order, we prove eventual uniqueness of the carousel base for the expected path and cycle counts in both blow-up models, and establish strict monotonicity of the regular-block constants.

1. Introduction

A tournament is an orientation of a complete graph. Write $H(T)$ for the number of directed Hamiltonian paths in a tournament T, and $C(T)$ for the number of directed Hamiltonian cycles. Paths are counted as vertex orderings, whereas cyclic starting points are identified for cycles. Let

$$P(n) = \max_{|V(T)|=n} H(T), \qquad \mu_n = \frac{n!}{2^{n-1}}, \qquad \nu_n = \frac{(n-1)!}{2^n}.$$

Szele [10] observed that a uniform random tournament has μ_n Hamiltonian paths in expectation: each vertex ordering is directed with probability $2^{-(n-1)}$. Hence $P(n) \geq \mu_n$. The corresponding expected cycle count is ν_n for $n \geq 3$. These expectations provide the natural normalization for the extremal problem.

Adler, Alon and Ross [9] improved the random lower bound by choosing an almost complete packing of edge-disjoint triangles and orienting each triangle cyclically, independently of the others. A vertex ordering containing one of these triangles in three consecutive positions is more likely to form a directed path than it is in the independent-edge model. For a uniformly random vertex ordering, the number of such occurrences converges in distribution to Poisson(1), from which they obtained the lower bound $(e-o(1))\mu_n$. Wormald [6] used a different construction: the directions between large vertex classes are prescribed by a fixed tournament, while the tournaments within the classes are random. His calculation for carousel bases gives a cycle lower constant exceeding 2.85588 along an infinite sequence of orders, already close to the constant considered below.

The upper bounds developed through permanent estimates. If A is the adjacency matrix of a tournament, then per A counts its spanning unions of vertex-disjoint directed cycles. In particular, it bounds the number of Hamiltonian cycles. Alon [2] applied Brégman's inequality to this permanent and optimized the resulting expression over the outdegrees, obtaining $P(n) = O(n^{3/2}\mu_n)$. Friedgut and Kahn [7] used an entropy argument to improve the polynomial exponent; their cycle bound is $O(n^{\alpha}\nu_n)$ with $\alpha \approx 1.249$. These estimates identify the correct exponential scale, but leave a polynomial gap between the upper and lower bounds. At the scale of a leading constant, one must retain information that a bound in terms of the degrees alone discards.

Divoux and Wang [12] determined the sharp leading constant. They proved that the maximum cycle count is $(L + o(1))\nu_n$ and the maximum path count is $(L + o(1))\mu_n$, where $L = \cosh 1 / \cos 1$. Their

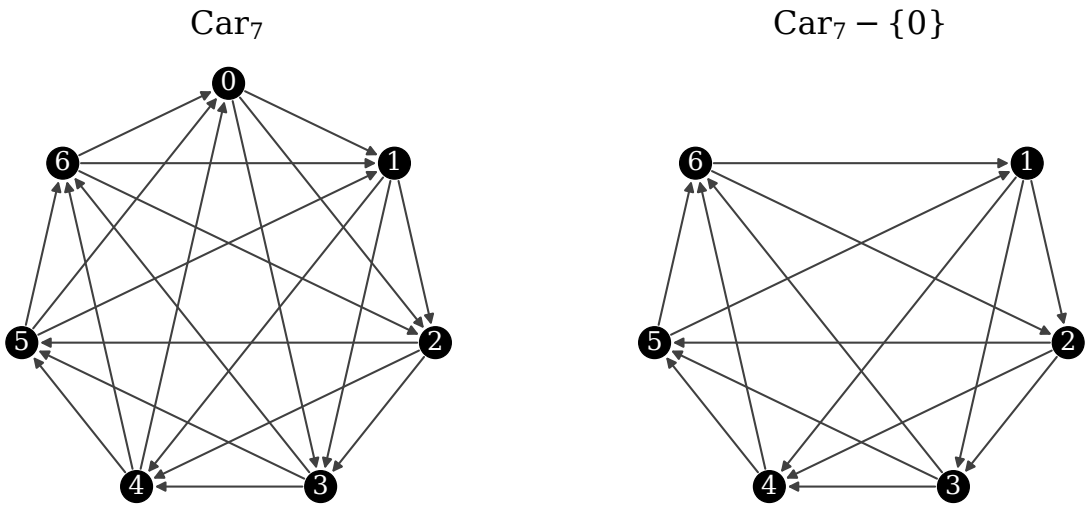


Figure 1: The carousel Car_7 and its even-order subtournament obtained by deleting vertex 0. Every vertex on the left has outdegree 3; those on the right have outdegree 2 or 3. All arcs are shown.

proof still uses permanents, but replaces a degree bound by a determinant approximation for nearly doubly stochastic matrices. After separating vertices with large degree imbalance, they classify a Hamiltonian cycle by the directed path segments that it induces on the exceptional set. These segments are encoded in modified adjacency matrices of the remaining tournament. Sinkhorn scaling and a determinant–permanent inclusion-exclusion formula then reduce the count to a ratio of determinants of a skew-symmetric matrix. A matrix comparison of Fang and Huang [11] bounds this ratio by its value for a transitive tournament. Odd carousels, and their one-vertex deletions at even orders, attain the bound asymptotically; a separate argument transfers the cycle result to paths.

The first author's path-counting preprint [14], publicly archived on 8 October 2026, had already established $(L-O(n^{-1}))\mu_n \le P(n) \le (C_*+O(n^{-1}))\mu_n$ and the analytic framework used below. Thus it closed the polynomial gap, while leaving a gap between the two leading constants $L < C_*$. That archive contains the positive path convolution, uniform permanent estimate, nonprincipal scaling and exceptional-vertex analysis; it predates the arXiv v1 submission of [12] on 16 October 2026. The present sharp estimate combines that framework with the spectral moment comparison of Fang and Huang [11]. It refines the leading asymptotic in [12, Theorem 1.4] and retains a uniform $O(n^{-1})$ normalized error through deletion, scaling and summation. We make no claim that this error order is optimal.

Theorem 1.1. There are absolute constants K', n_1 such that every integer $n \ge n_1$ satisfies

$$(L - K'/n)\mu_n \le P(n) \le (L + K'/n)\mu_n.$$

For odd n, the carousel Car_n has vertex set $\mathbb{Z}/n\mathbb{Z}$, with $i \to j$ precisely when the representative of $j - i$ modulo n belongs to $\{1, \dots, (n-1)/2\}$. The lower bound is attained, up to an error of $O(\mu_n/n)$, by odd carousels and by the even tournaments obtained by deleting one vertex from Car_{n+1}.

The second part of the paper concerns Wormald's random blow-up models. Fix an odd integer $d \ge 3$ and a tournament T^* on d vertices. Each vertex is replaced by a block of m vertices, and the directions between blocks are prescribed by T^*. In $\mathcal{T}(T^*, m)$ the internal tournaments are independent and uniform; in $\mathcal{R}(T^*, m)$ they are independent and uniform among regular tournaments, so m must be odd. Put $N = dm$. The expected cycle and path counts are normalized by ν_N and μ_N, respectively.

For a regular base, Theorem 4 of Wormald [6] expresses the limiting constant in terms of an angular Hessian, subject to its determinant being nonzero. He computes this determinant for several bases and con-

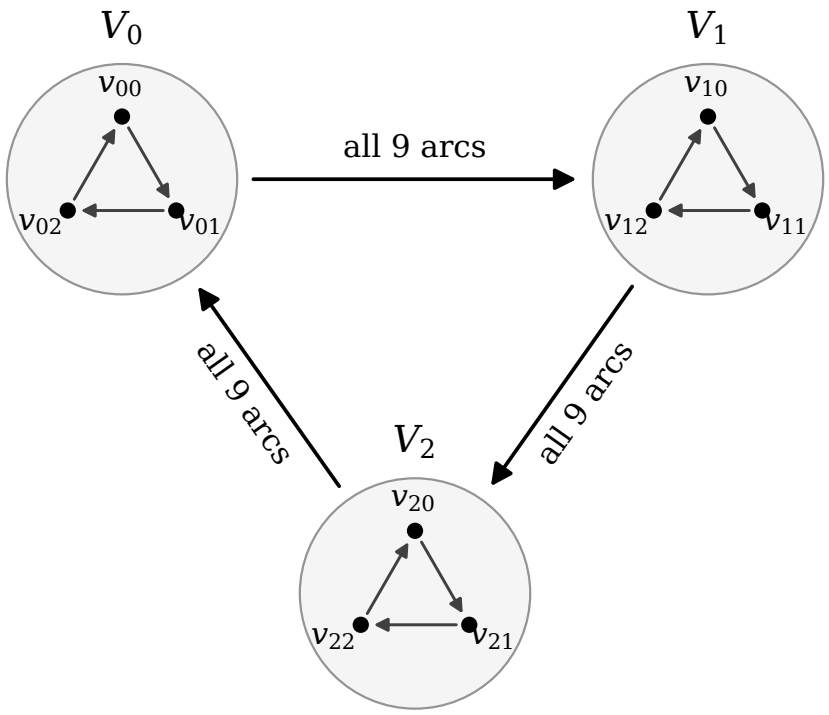


Figure 2: A blow-up of the directed triangle. Each internal block shown is a directed triangle; a thick arrow between blocks represents all 9 arcs in that direction. In the random models, the internal blocks are chosen independently from the specified distributions.

jectures that the condition always holds. We prove positive definiteness and evaluate the determinant for every regular base.

Theorem 1.2. Let T^* be regular, with adjacency matrix A. Set $S = A - A^T$, $C = S/d$, and $E = (e_1 - e_d, \dots, e_{d-1} - e_d)$. The Hessian $\mathbf{H}$ of Wormald's angular function at the origin is positive definite, and

$$\mathbf{H} = E^T(I + C^2)(I - C^2)^{-1}E, \qquad \mathcal{H} = d\,\frac{\det(I + C^2)}{\det(I - C^2)} > 0,$$

$$\sqrt{d/\mathcal{H}} = \rho_d(S) := \frac{\det(I + S/d)}{\det(I + iS/d)} \in \mathbb{Q}_{>0}.$$

Here $\mathcal{H} = \det \mathbf{H}$, and the square root is positive. The angular function is defined in Section 9. Together with Theorems 1, 4 and 5 of Wormald [6], this shows that, for fixed d, the normalized expected cycle and path counts in $\mathcal{T}(T^*, m)$ both tend to $\rho_d(S)$, while the corresponding limits in $\mathcal{R}(T^*, m)$ are $e^{1/d}\rho_d(S)$.

The model constants can also be obtained directly from Divoux and Wang [12, Proposition 5.1], without the Hessian calculation; Section 10.1 records that derivation. Their Proposition 7.1 and Lemma 9.1 already give the spectral maximum and the carousel value. The next corollary combines these results with the equality conditions of Fang and Huang [11], the zero-row-sum condition for a regular base, and a separate bound for nonregular bases.

Corollary 1.3. For each fixed odd $d \geq 3$, there exists M_d such that, for every $m \geq M_d$, the carousel Car_d is the unique base, up to isomorphism, maximizing the expected number of Hamiltonian cycles and paths in both models. For $\mathcal{R}(T^*, m)$, only odd m are considered.

The closed form below is the binomial evaluation in Divoux and Wang [12, Section 7 and Lemma 9.1], with the factor $e^{1/d}$ supplied by Wormald [6, Theorem 1]. The known frequencies in Fang and Huang [11, Lemma 4.2] give the following further consequence.

Corollary 1.4. Set

$$r_d = \frac{(1+1/d)^d + (1-1/d)^d}{(1+i/d)^d + (1-i/d)^d}, \qquad \mathfrak{r}_d = e^{1/d} r_d.$$

Along odd integers $d \geq 3$, the sequence $\mathfrak{r}_d$ increases strictly to L.

Theorem 1.2 directly proves Wormald's Hessian conjecture [6, Conjecture 6]. Corollary 1.3 combines existing counting and spectral results with the equality and nonregular-base arguments given here. The limit in Conjecture 2 follows from the closed form in [12] and Wormald's factor $e^{1/d}$; Corollary 1.4 adds strict monotonicity. Section 10 retains the expansion through d^{-5}, with remainder $O(d^{-6})$, as a calculation from that closed form.

Bhattacharjee [13, Theorems 1.1 and 1.3, Proposition 3.1, Lemma 3.2 and Corollary 3.3] also treats Wormald's Conjectures 2 and 6. The original angular Hessian, positivity, determinant/rational-square formulas and carousel limit were obtained independently in our Wormald project [15] and in his work. We retain our direct derivation and record the overlap. Our October 8 path archive [14] concerns a different part of the paper and is not used to date the Hessian result. Bhattacharjee's public release dated 14 October [16] already contains the full Hessian; the fixed Liu–Ye manuscript revision cited in [15] is dated 19 October. Git commit dates and public-release dates have different meanings, and these records do not establish priority for our Hessian calculation.

The contribution and dependency structure is as follows.

Component	Scope and relation to prior work
Constant-factor path framework	Independently archived in [14]; the upper constant there was $C_* > L$.
Sharp global path estimate	Uniform final $O(n^{-1})$ error, obtained by combining the counting estimates with [11]; the sharp leading asymptotic is in [12].
Signed permanent estimate	The independent proof from [14] is retained; the counting applications use its nonnegative specialization.
Angular Hessian and carousel limit	Independent derivation overlapping [13,16]; no priority claim is based on the earlier path archive.
Fixed-base extremality	Equality analysis, nonregular-base decay and a finite-class comparison complete eventual uniqueness for expectations.
Regular-block monotonicity	A strict monotonicity consequence of the known carousel spectrum.

For Corollary 1.3 the order d is fixed first. There are finitely many base isomorphism classes; regular noncarousel bases have a strict limiting gap, and the normalized expectations of nonregular bases tend to zero. A common threshold can therefore be chosen for all four expectations and all bases of that fixed order. The statement concerns expectations, and the regular-block model only allows odd m.

1.1 The counting estimates and their relation to earlier methods

McCullagh [5] obtained the determinant approximation with an $O(n^{-1})$ relative error under moderate-deviation assumptions. Divoux and Wang [12, Proposition 3.1] prove a uniform $1 + o(1)$ version for nonnegative doubly stochastic matrices; their coefficient estimates and Gamma averaging can be refined to retain $O(n^{-1})$ in that setting. Section 3 gives the form needed here, with fixed entry and operator-norm bounds, and includes the detailed proof. The quantitative contribution of Theorem 1.1 is the final path

count after deletion, scaling and summation; the determinant main term and earlier $O(n^{-1})$ approximations provide its analytic background.

The proof expands the permanent by imposing distinct row and column indices separately. Double centering eliminates terms with a vertex of degree one. Components in which every vertex has degree two give Gaussian pairings and hence the determinant main term; the other components are estimated after contracting their degree-two chains. The resulting coefficient estimates hold over a range growing with the matrix order. For the tail, permanent polarization reduces the required complex-matrix bound to a positive semidefinite one. The complex Gaussian identity and arithmetic–geometric mean estimate used by Han and Niles-Weed [3, Lemmas 4.3–4.4] provide this bound; we then apply Cauchy's coefficient estimate. Together, the coefficient estimates and the tail bound give the uniform $O(n^{-1})$ error.

For paths, we use the identity

$$H(T) = \sum_{U \subseteq V(T)} \det(I + A[U]) \operatorname{per} A[U^c].$$

This is a specialization of the determinant–permanent identities discussed by Irving and Omar [1]. All its summands are nonnegative for a tournament. We can therefore estimate the permanent in each term without controlling cancellation in the final sum. The terms with large U have a sufficiently small factorial tail. For the remaining terms, the task is to apply the permanent approximation with an error uniform in both the tournament and the deletion set. We also retain the weights associated with each deletion-set size: their fixed moments are summable. Replacing every size by the largest permitted value would introduce unnecessary logarithmic losses.

The exceptional vertices require a further estimate. We split the adjacency matrix into four blocks according to the exceptional set and its complement, and expand the permanent over the matchings between these blocks. A cross-block matching removes core rows and core columns that need not have the same indices. Thus estimates restricted to principal submatrices do not suffice. Section 4 treats these nonprincipal submatrices, performs the diagonal scaling, and records the change in the Gaussian determinant factor. This four-block expansion is the way exceptional vertices are counted here; the directed-segment classification in [12] concerns the cycle count.

It remains to ensure that these estimates give the claimed error for every tournament. We may assume $H(T) \geq \mu_n$. Brégman's inequality first bounds the sum of squared degree imbalances. Tournaments with a sufficiently large value of this sum are bounded directly; those with an exceptional vertex are handled by the cross-block expansion. For the remaining tournaments, the scaling calculation retains an exponential penalty for degree imbalance. Some intermediate errors have square-root size, which would be too large if estimated in isolation. Together with the penalty, they contribute only $O(n^{-1})$ to the upper bound. The main term is

$$\rho_n(S) = \frac{\det(I + S/n)}{\det(I + iS/n)}, \qquad S = A - A^T.$$

As in [12], the comparison of Fang and Huang [11] bounds this ratio by its value at the sign matrix of a transitive tournament. That value is $L + O(n^{-1})$. For the lower bound, the two-sided path estimate applies to odd carousels and to their even-order one-vertex deletions: the degree imbalances are respectively zero and of absolute value one. These arguments give Theorem 1.1 directly. Proposition 6.6 supplies the all-order spectral comparison; Proposition 7.3 isolates the uniform score penalty, and Corollary 8.2a gives the resulting degree-variance stability estimate.

1.2 Wormald's models

Wormald [6] obtains the fixed-base counting formula by projecting a Hamiltonian cycle onto a closed walk in the base tournament. Inside each block, the cycle induces a collection of directed paths. Generating functions for these walks and paths lead to a multivariate coefficient problem, to which smooth-point asymptotics apply.

For a regular base, the remaining constant involves the determinant of an angular Hessian defined through an implicit equation. This leaves both a nondegeneracy question and a comparison problem between different bases.

We differentiate the original implicit equation to obtain Theorem 1.2, including positive definiteness of the full Hessian. This supplies an unconditional justification for Wormald's nondegeneracy assumption. The identity $\sqrt{d/\det \mathbf{H}} = \rho_d(S)$ also links the Hessian to the existing spectral theory. The model limit $\rho_d(S)$ itself is already a direct application of [12, Proposition 5.1]; using Wormald's conditional Hessian formula to infer nondegeneracy would not be a valid substitute for the calculation.

For regular bases, we supplement the comparison in [12] with the equality classification from Fang and Huang [11]. Signed similarity together with zero row sums forces the carousel. Nonregular bases are bounded separately by Brégman's inequality. Since only finitely many bases occur at fixed order, the strict limiting gaps give Corollary 1.3. The short spectral proof gives Corollary 1.4; the separate expansion records higher-order terms. The base order is fixed before the block order tends to infinity.

1.3 Further questions and organization

The path maximizers at each finite order and the coefficient of the first correction term remain open problems. It would also be useful to estimate M_d in terms of d. Allowing the base order and the block order to grow together would require error bounds uniform in the base order, beyond the fixed-base asymptotics used here.

Section 2 fixes notation. Sections 3 and 4 give the permanent and nonprincipal scaling estimates; Section 5 records the positive path convolution, summable weights and uniform tail. Section 6 proves the spectral comparison for all orders. Sections 7 and 8 establish the sharp path bound directly, followed by degree-variance stability. Sections 9–11 treat the original angular Hessian and fixed-base model extrema. The detailed proof companion retains the original section and equation numbers for the analytic estimates and auxiliary algebra. This concise main text states their precise hypotheses and develops the counting and model arguments; the companion supplies the longer technical derivations.

2. Preliminaries and notation

2.1 The random baseline and error conventions

Let $\mathfrak{S}_n$ be the symmetric group on $[n]$, and let I_π indicate that the consecutive edges of π all point forward. Each ordering uses $n-1$ distinct undirected edges, so

$$H(T) = \sum_{\pi \in \mathfrak{S}_n} I_\pi(T), \qquad \mathbb{E}I_\pi = \prod_{j=1}^{n-1} \frac{1}{2} = 2^{-(n-1)},$$

$$\mathbb{E}H(T) = \sum_{\pi \in \mathfrak{S}_n} \mathbb{E}I_\pi = n!2^{-(n-1)} = \mu_n.$$

The maximum is at least the mean, so $P(n) \geq \mu_n$. Let $\mathcal{T}_n$ denote all tournaments with vertex labels $[n]$. In this paper, a uniform $O(n^{-1})$ error means

$$\exists K < \infty \; \exists n_0 \; \forall n \geq n_0 \; \forall T \in \mathcal{T}_n : \quad |\text{error}(n,T)| \leq K/n.$$

The constants K, n_0 are uniform over tournaments and deletion sets. They may depend on additional fixed parameters b, A_0, B_0, when present.

2.2 Sign matrices and spectral factors

Definition 2.1 (Sign matrices and scores).

The adjacency matrix A has zero diagonal and $A_{ij} = 1$ precisely when $i \to j$. Define

$$S = A - A^{\mathsf{T}}, \qquad 2A = J - I + S, \qquad s = S\mathbf{1}.$$

Here I is the identity matrix in the current dimension, $J = \mathbf{1}\mathbf{1}^{\mathsf{T}}$, and $\mathbf{1}$ is the all-ones column vector. In $I_p, J_p, \mathbf{1}_p$, the subscript explicitly specifies the dimension p. The superscript T denotes transpose. If d_i^+ is the outdegree, then $s_i = 2d_i^+ - (n-1)$ and $\sum_i s_i = 0$. The Euclidean and maximum vector norms are

$$\|x\|_2 = \left(\sum_i |x_i|^2\right)^{1/2}, \quad \|x\|_\infty = \max_i |x_i|.$$

We use the notation

$$\mathcal{E} = \|s\|_2^2, \qquad d = \|s\|_\infty, \qquad a = \frac{s}{n-1}, \qquad \tau = \|a\|_2^2.$$

The normalized scores a and τ are defined for $n \geq 2$. An odd-order tournament is regular when every vertex has outdegree $(n-1)/2$, equivalently $s = 0$. An even-order tournament is balanced when every outdegree is $n/2$ or $n/2 - 1$, equivalently $s_i \in \{-1, 1\}$. Thus d measures the largest vertex imbalance, whereas τ measures the total squared imbalance after normalization.

Definition 2.2 (Submatrices and permanents).

For a vertex set U, the notation $A[U] = A[U, U]$ denotes the principal submatrix. For possibly different row and column sets R, C, the submatrix $A[R, C]$ selects those rows and columns, always in increasing original vertex-label order. The same convention applies to a general matrix $M[R, C]$. The complement U^c is taken in the current full vertex set. Empty sums equal zero and empty products equal one; both the determinant and the permanent of the empty matrix equal one. For a square matrix of order m, denoted by M, the permanent is

$$\operatorname{per} M = \sum_{\pi \in \mathfrak{S}_m} \prod_{i=1}^{m} M_{i,\pi(i)}.$$

Definition 2.3 (Frequencies and spectral factors).

The singular values of a real matrix M are the nonnegative square roots of the eigenvalues of $M^{\mathsf{T}}M$. The operator norm $\|M\|_{\mathrm{op}} = \sup_{\|x\|_2=1} \|Mx\|_2$ equals the largest singular value. The square of the Frobenius norm $\|M\|_F = (\sum_{i,j} |M_{ij}|^2)^{1/2}$ is the sum of the squared singular values. For the skew-symmetric matrix S, write $\pm i\lambda_j$ for its nonzero eigenvalue pairs, where $i^2 = -1$ and $\lambda_j > 0$, adding zero frequencies when needed. Each λ_j occurs twice among the singular values. Define

$$x_j = \frac{\lambda_j^2}{n^2}, \qquad D_n(S) = \det(I - S^{\mathsf{T}}S/n^2)^{-1/2},$$

$$\rho_n(S) = D_n(S) \det(I + S/n) = \prod_j \frac{1 + x_j}{1 - x_j}.$$

The skew-symmetric singular values occur in pairs, and $\|S\|_F^2 = n(n-1)$, so

$$\sum_j x_j = \frac{n-1}{2n}, \qquad \|S/n\|_{\mathrm{op}}^2 \leq \frac{n-1}{2n} < \frac{1}{2}.$$

Therefore D_n and ρ_n are positive and bounded by absolute constants. In particular,

$$D_n(S) = \det(I + iS/n)^{-1}.$$

For every i,

$$s_i = \sum_{j\neq i}(A_{ij} - A_{ji}) = d_i^+ - [(n-1) - d_i^+] = 2d_i^+ - (n-1),$$

$$\sum_i s_i = \sum_{i<j}(S_{ij} + S_{ji}) = 0, \qquad \|S\|_F^2 = \sum_{i\neq j} 1 = n(n-1).$$

In a real orthonormal basis, a real skew-symmetric matrix becomes a direct sum of blocks $\left(\begin{smallmatrix} 0 & \lambda_j \\ -\lambda_j & 0 \end{smallmatrix}\right)$ and zero blocks. Consequently,

$$\begin{aligned}
\det(I + S/n) &= \prod_j (1 + \lambda_j^2/n^2) = \prod_j (1 + x_j), \\
\det(I - S^{\mathsf{T}}S/n^2) &= \prod_j (1 - x_j)^2, \\
D_n(S) &= \prod_j (1 - x_j)^{-1}, \\
\det(I + iS/n) &= \prod_j (1 - \lambda_j/n)(1 + \lambda_j/n) = \prod_j (1 - x_j), \\
\rho_n(S) &= \prod_j \frac{1 + x_j}{1 - x_j}.
\end{aligned}$$

Each frequency occurs twice in the squared Frobenius norm, so

$$2\sum_j \lambda_j^2 = n(n-1), \qquad \sum_j x_j = \frac{n-1}{2n}, \qquad \|S/n\|_{\mathrm{op}}^2 \le \sum_j x_j < \frac{1}{2}.$$

For $n = 1$, we have $S = [0]$, the frequency sum is empty, and $D_1(S) = \rho_1(S) = 1$.

Every log below is a natural logarithm. Using $-\log(1-x) = \int_0^x (1-u)^{-1}\,du \le 2x$ for $0 \le x \le 1/2$, we obtain

$$0 \le \log D_n(S) \le 2\sum_j x_j < 1, \qquad 1 \le D_n(S) < e.$$

Similarly, $\log(1+x) \le x$ gives $\log \rho_n(S) < 3/2$. Multiplication by $D_n(S)$ or $\rho_n(S)$ therefore preserves the order of the error. Lemma 6.1 gives a sharper bound.

3. A uniform permanent approximation

The counting argument needs a permanent estimate uniform under scaling and logarithmically many independent row and column deletions. We state the analytic theorem with its exact parameters; its full coefficient proof is in **Detailed proof companion, §§3.2–3.6**, retaining the original numbering. This section gives the proof structure and the estimates that close it. The supplement forms an integral part of the proof.

McCullagh [5] obtained the determinant main term under moderate-deviation assumptions. The argument retained here comes from the earlier path project [14]. It allows signed inputs, although Sections 4–8 use only the nonnegative case. Divoux–Wang [12] give a related nonnegative theorem with an $o(1)$ error; extracting the $O(n^{-1})$ rate uses the uniform coefficient and tail estimates recorded below.

Theorem 3.1 (Uniform permanent approximation).

Fix $0 \le C < \infty$ and $0 \le q < 1$. Let $n \ge 1$, and suppose $E \in \mathbb{R}^{n\times n}$ satisfies

$$E\mathbf{1} = E^{\mathsf{T}}\mathbf{1} = 0, \qquad \max_{i,j} |E_{ij}| \le C, \qquad \|E/n\|_{\mathrm{op}} \le q.$$

Put $B = E/n$. Uniformly over all matrices satisfying these conditions,

$$\frac{\operatorname{per}(J_n + E)}{n!} = \det(I - BB^{\mathsf{T}})^{-1/2} + O_{C,q}(n^{-1}). \tag{3.1}$$

Equation (3.1) can also be written as the determinant factor times the relative error $1 + O_{C,q}(n^{-1})$. The subscript C, q means that the error constant and the threshold dimension may depend on these two parameters fixed in advance, but must apply simultaneously to every admissible matrix.

Proof with the coefficient estimates delegated to the supplement. Choose fixed $1 < \sigma < R < 1/q$ (omit the upper restriction if $q = 0$), and put

$$f(z) = \frac{\operatorname{per}(J_n + zE)}{n!}, \qquad G(z) = \det(I - z^2 BB^{\mathsf{T}})^{-1/2}, \qquad \Gamma(r) = \exp\!\left(\frac{C^2 r^2}{2(1 - q^2 r^2)}\right).$$

Since $\|B\|_F^2 \le C^2$, $1 \le G(1) \le \Gamma(1)$ and $|G(z)| \le \Gamma(|z|)$ on the fixed disk. The positive square-root branch is determined at zero.

Separate inclusion–exclusion for row and column indices expresses coefficients as bipartite multigraph sums. Double centering eliminates degree-one vertices. Degree-two components are exactly the Gaussian pairings and sum to G. For the remaining components, contract degree-two chains **before taking absolute values**: the alternating products of B, B^{T} retain the spectral gap. A chain of length $\ell \ge 2$ has contraction bounded by $C^2 q^{\ell-2}/n$. The residual core is classified by excess (edges minus vertices), giving one factor n^{-1} per excess. The exact labeling compensation and the core activity bound are proved in supplement §§3.2–3.3. This order of contraction preserves the operator-norm control.

For some fixed $\alpha > 0$, set $M = \lfloor \alpha n \rfloor$. The finite coefficient window, including restoration of $n^k/(n)_k$, has error $K_1/n + K_2/n^2$ (supplement §3.4). Its deviation is of order k^2/n, and the Gaussian coefficients are summed with a fixed radius greater than one to absorb this weight. Permanent polarization and the positive-semidefinite permanent estimate give $\max_{|z|=R} |f(z)| \le \exp(\beta\sqrt{n})$, up to a fixed factor, where $\beta = RC/(1 - Rq)$; Cauchy's estimate then controls the coefficients above M (supplement §3.5). Together these bounds yield

$$|f(1) - G(1)| \le \frac{K_1}{n} + \frac{K_2}{n^2} + K_3 e^{\beta\sqrt{n} - \alpha n \log R} + K_4 e^{-\alpha n \log \sigma}.$$

Every constant and the threshold depend only on C, q and the radii fixed from them. Both exponential terms are $o(n^{-1})$. Finitely many smaller orders are covered using $|f(1)| \le (1 + C)^n$ and $G(1) \le \Gamma(1)$. This proves the uniform absolute bound; $G(1) \ge 1$ gives the relative form. □

The supplement also records the signed example with fixed parameters at arbitrarily large orders and the two estimates needed to quantify the nonnegative proof of [12]. The applications use the stated nonnegative specialization.

4. Scaling and nonprincipal submatrices

Write $a = S\mathbf{1}/(n-1)$, $\tau = \sum a_i^2$, $P_p = J_p/p$ and $\Pi_p = I_p - P_p$. For a matrix X of order p, let $\mathfrak{m}(X) = \sum X_{ij}$ and $g(X) = (X\mathbf{1} - \mathbf{1}, X^{\mathsf{T}}\mathbf{1} - \mathbf{1})$. The normalization $C = 2A/(n-1)$ has marginals $\mathbf{1} + a$ and $\mathbf{1} - a$. The three estimates below control balancing, its logarithmic cost, and the Gaussian factor. Their parameter dependence is the input to the uniform bounds in Lemmas 4.4–4.5.

4.1 Scaling and Gaussian factors

Lemma 4.1 (Local scaling).

Let $p \ge 1$, and let X have total mass p and be a real $p \times p$ matrix. Write

$$\alpha = X\mathbf{1}_p - \mathbf{1}_p, \qquad \beta = X^{\mathsf{T}}\mathbf{1}_p - \mathbf{1}_p, \qquad \varepsilon = \max(\|\alpha\|_\infty, \|\beta\|_\infty).$$

Assume $|X_{ij}| \le C/p$, that the sum of absolute values in each row and each column is at most 2, and that

$$X_0 = X - \frac{\alpha\mathbf{1}_p^{\mathsf{T}} + \mathbf{1}_p\beta^{\mathsf{T}}}{p}, \qquad E = X_0 - P_p$$

satisfies $|E_{ij}| \le C_0/p$ and $\|E\|_{\mathrm{op}} \le q < 1$, where $C, C_0 \ge 0$ and $0 \le q < 1$ are fixed parameters. Define

$$L_0 = \frac{3}{2} + C_0 + \frac{C_0^2}{1-q},$$

$$\varepsilon_0 = \min\left\{\frac{1}{256L_0^2}, \frac{1-q}{2(32L_0+2)}\right\}.$$

If $\varepsilon \le \varepsilon_0$, the condition $\sum_i x_i = \sum_j y_j$ is satisfied by finite real potential vectors x, y such that every row sum and column sum of $B = \mathrm{diag}(e^x)X\,\mathrm{diag}(e^y)$ equals 1. Moreover,

$$\|(x,y)\|_\infty \le 4L_0\varepsilon, \qquad |B_{ij}| \le 2C/p, \qquad \|B - P_p\|_{\mathrm{op}} \le (1+q)/2.$$

If X is entrywise nonnegative, then B is doubly stochastic and has the same zero pattern as X.

Lemma 4.2 (Displacement and capacity).

Under the preceding hypotheses, assume in addition that $X \ge 0$. Let $\theta_X = \sum_i x_i + \sum_j y_j$ denote the total scaling potential. There is a constant depending only on the fixed density and spectral-gap parameters such that

$$\|(x,y)\|_2 \le K\|(\alpha,\beta)\|_2, \qquad \|B - X\|_F \le \frac{K}{\sqrt{p}}\|(\alpha,\beta)\|_2,$$

$$0 \le \theta_X \le K\|(\alpha,\beta)\|_2^2.$$

General mass. For a nonnegative matrix X of arbitrary positive total mass, suppose that $\widetilde{X} = pX/\mathfrak{m}(X)$ satisfies the hypotheses of Lemma 4.1, and apply the preceding conclusions to $\widetilde{X}$. If B is its scaling, the following exact identities hold:

$$\theta_X = \theta_{\widetilde{X}} + p\log\frac{p}{\mathfrak{m}(X)}, \qquad \operatorname{per} X = e^{-\theta_X}\operatorname{per} B.$$

In particular, if $\kappa = \mathfrak{m}(X) - p$, then $\operatorname{per} X \le e^{\kappa}\operatorname{per} B$. When the mass differs from p, θ_X itself need not be nonnegative.

Proof mechanism. Fix the gauge $\sum x_i = \sum y_j$. The reference matrix H_0 has diagonal blocks I and off-diagonal blocks X_0, X_0^{T}, while the actual linearization H_X has diagonal blocks $\mathrm{diag}(X\mathbf{1}), \mathrm{diag}(X^{\mathsf{T}}\mathbf{1})$ and off-diagonal blocks X, X^{T}. Put $u = (\mathbf{1}, -\mathbf{1})$ and $N = uu^{\mathsf{T}}/(2p)$ to fix the gauge. On the zero-sum subspaces, the inverse of $\mathsf{H}_0 + N$ is expressed through $(I - EE^{\mathsf{T}})^{-1}$ and $(I - E^{\mathsf{T}}E)^{-1}$. The dense entry bound and operator gap give its induced 1- and ∞-norms at most L_0. Both corresponding norms of $\mathsf{H}_X - \mathsf{H}_0$ are at most 3ε; hence the Neumann series bounds the inverse of $\mathsf{H}_X + N$ by $2L_0$. The stated threshold then makes the nonlinear map a contraction in the radius $4L_0\varepsilon$ ball. This proves existence and the stated entry and spectral bounds. The block inverse and contraction estimates are supplied in Detailed proof companion §4.1.

For nonnegative X, apply this construction to the convex potential

$$\Phi_X(x,y) = \sum_{ij} X_{ij}e^{x_i+y_j} - \sum x_i - \sum y_j.$$

Along the segment to the balanced point, its Hessian on the gauge subspace has a positive lower bound depending only on the fixed parameters. Integration gives the two-norm bound, and the entry bound gives the Frobenius displacement. Comparing the potential at zero and at its minimum gives $0 \le \theta_X \le K\|g(X)\|_2^2$. Scalar normalization proves the general-mass identity and $-\theta_X \le \kappa$. In particular, the nonnegativity of the potential cost is used at total mass p; the general-mass argument uses the one-sided bound $-\theta_X \le \kappa$.

Define $\mathcal{G}(Z) = \det(I - Z^{\mathsf{T}}Z)^{-1/2}$ for $\|Z\|_{\mathrm{op}} < 1$; zero padding and changing to ZZ^{T} preserve its value.

Lemma 4.3 (Gram deletion and stability).

Suppose $\|Z\|_{\mathrm{op}} \le q_* < 1$. Delete the row set I and the column set J, leaving the sets R, T. Then

$$0 \le \log \mathcal{G}(Z) - \log \mathcal{G}(Z[R,T]) \le \frac{\sum_{i\in I}\|Z_{i,\cdot}\|_2^2 + \sum_{j\in J}\|Z_{\cdot,j}\|_2^2}{2(1-q_*^2)}.$$

Double centering. At order $m \ge 1$, assume $\|W\|_{\mathrm{op}} \le q_*$ for a square retained matrix W. Set $u = \mathbf{1}_m/\sqrt{m}$. Then

$$0 \le \log \mathcal{G}(W) - \log \mathcal{G}(\Pi_m W \Pi_m) \le \frac{\|W^{\mathsf{T}}u\|_2^2 + \|Wu\|_2^2}{2(1-q_*^2)}.$$

Stability. For two real matrices of the same dimensions whose operator norms are both at most q_*, we also have

$$|\log \mathcal{G}(U) - \log \mathcal{G}(V)| \le \frac{\|U\|_F + \|V\|_F}{2(1-q_*^2)}\|U - V\|_F.$$

Proof. Integrate the derivative of $-\frac{1}{2}\log\det(I-H)$ along the positive-semidefinite Gram loss caused by deleting rows. Its trace is the sum of the deleted squared row norms. Switch Gram sides for column deletion and apply the same argument to the constant-direction projections. The stability inequality follows along the Gram segment from $\|U^{\mathsf{T}}U - V^{\mathsf{T}}V\|_* \le (\|U\|_F + \|V\|_F)\|U - V\|_F$. □

4.2 Uniform deletion bounds

Lemma 4.4 (Nonprincipal permanent bound).

Fix $0 \le b < 1$ and positive constants A_0, B_0. For a tournament core of order $N \ge 2$, set $a = S\mathbf{1}/(N-1)$ and $\tau = \sum_i a_i^2$. Let $I, J \subseteq [N]$ be the deleted row and column sets, respectively. Here I, J are sets, distinguished from the identity and all-ones matrices by the set operations below. Assume

$$\max_i |a_i| \le b, \qquad \tau \le A_0 \log N, \qquad |I| = |J| = t \le B_0 \log N.$$

Define $\ell_i = (1+a_i)^{-1}$, $r_i = (1-a_i)^{-1}$, and

$$\Gamma = \prod_i (1 - a_i^2), \qquad \mathcal{M}_\tau = \frac{\tau + \tau^2}{N} + \frac{\tau^{3/2}}{\sqrt{N}}.$$

Let $R = [N] \setminus I$ and $T = [N] \setminus J$ be the retained row and column sets, respectively; here T is a column index set. Then

$$\operatorname{per} A[R,T] \le e^{-1} D_N(S) \Gamma \left(\prod_{i\in I}\ell_i\right)\left(\prod_{j\in J} r_j\right)\frac{(N-t)!}{2^{N-t}}\exp(\varepsilon_{\tau,t}),$$

$$\varepsilon_{\tau,t} \le K\left[\mathcal{M}_\tau + \sqrt{\tau/N} + \frac{t+1}{N} + t\sqrt{\tau/N} + \frac{t^2}{N}\right].$$

For every sufficiently large N, this estimate holds simultaneously for every admissible tournament and every pair I, J. The constant K and the threshold depend only on b, A_0, B_0.

Proof. Use $n = N$ and $m = n - t$. Perform paired score correction and mass normalization:

$$C = \frac{2A}{n-1}, \qquad C' = \operatorname{diag}(\ell) C \operatorname{diag}(r),$$
$$\nu = \mathfrak{m}(C') - n, \quad \widehat{C} = \frac{nC'}{n+\nu},$$
$$X = \frac{n}{m}\widehat{C}[R,T], \quad \kappa = \mathfrak{m}(X) - m.$$

Put $v_i = a_i/(1-a_i^2)$, $w_i = a_i^2/(1-a_i^2)$, $V = \sum v_i$, $W = \sum w_i$. The cancellation of the first-order mass change is recorded by the exact identity

$$\nu = \frac{W^2 - V^2 + W + 2w^{\mathsf{T}} S v}{n-1}, \qquad |\nu| \le K\mathcal{M}_\tau.$$

If e, f are the marginal errors of $\widehat{C}$, deleting the independent row and column sets gives

$$\kappa = \frac{n}{m}\left[-\sum_{i\in I} e_i - \sum_{j\in J} f_j + \sum_{i\in I, j\in J} \widehat{C}_{ij} - \frac{t^2}{n} \right]. \tag{4.1}$$

Normalize $\widetilde{X} = mX/(m+\kappa)$. The preprocessing estimates in Detailed proof companion §4.3 give, uniformly,

$$|\kappa| \le K(t\sqrt{\tau/n} + t^2/n), \quad \|g(\widetilde{X})\|_\infty \le K(\sqrt{\tau/n} + t/n), \quad \|g(\widetilde{X})\|_2 \le K(\sqrt{\tau} + t/\sqrt{n}).$$

The entries of $\widetilde{X}$ lie in $[0, K/m]$ and its doubly centered kernel has operator norm at most $1/\sqrt{2} + o(1)$. Fixing the gap $q = 3/4$ therefore makes Lemma 4.1 applicable above a common threshold. Its scaling B_X has entries $O(1/m)$, with centered norm eventually at most $4/5$. Lemmas 4.2–4.3 and the same preprocessing give

$$-\theta_X \le \kappa, \qquad \log \mathcal{G}(B_X - P_m) \le \log D_n(S) + K\big(\sqrt{\tau/n} + (t+1)/n\big).$$

The exact restoration is

$$\operatorname{per} A[R,T] = \Gamma \prod_{i\in I} \ell_i \prod_{j\in J} r_j \left[\frac{m(n-1)(n+\nu)}{2n^2}\right]^m e^{-\theta_X} \operatorname{per} B_X. \tag{4.2}$$

Theorem 3.1 gives $\operatorname{per} B_X = (m!/m^m)\mathcal{G}(B_X - P_m)(1+O(m^{-1}))$. The remaining scalar is $(m!/2^m)(1-1/n)^m(1+\nu/n)^m$, whose logarithmic deviation from $e^{-1}m!/2^m$ is $O((t+1)/n + |\nu|)$. Combining these bounds proves the displayed error exponent. All gaps, densities and thresholds were fixed from b, A_0, B_0, which establishes the stated uniform quantifiers. □

Lemma 4.5 (Permanent approximation for small scores).

Suppose $d = \|S\mathbf{1}_n\|_\infty = o(\sqrt{n})$. Fix $0 < B_0 < \infty$, delete any $t \le B_0 \log n$ rows and t columns, and write $m = n - t$. Uniformly over these choices,

$$\operatorname{per} A[R,T] = e^{-1} D_n(S) \frac{m!}{2^m}\left[1 + O_{B_0}\left(\frac{(d+t+1)^2}{n}\right)\right]. \tag{4.3}$$

More precisely, for fixed $B_0 > 0$ there are constants $C_{\mathrm{err}}, \delta > 0$ and an integer $N_0 \ge 4$ such that the following holds uniformly for $n \ge N_0$. Let S be a tournament sign matrix and let $I, J \subseteq [n]$ satisfy

$|I| = |J| = t \le B_0 \log n$. Put $d = \|S\mathbf{1}_n\|_\infty$ and $\eta = (d+t+1)^2/n$. If $\eta \le \delta$, the absolute relative error in equation (4.3) is at most $C_{\mathrm{err}}\eta$. Here $R = [n] \setminus I$, $T = [n] \setminus J$, and $m = n - t$; the two deleted sets may be chosen independently. In the asymptotic range of the lemma, $\eta \to 0$.

Proof. Take $X = (n/m)C[R,T]$ directly, without paired score correction. Equation (4.1) and the marginal identities give

$$|\kappa| = O(t(d+t)/n), \quad \|g(X)\|_\infty = O((d+t)/n), \quad \|g(X)\|_2 = O((d+t)/\sqrt{n}).$$

After normalizing to mass m, Lemmas 4.1–4.2 give $|\theta_X| = O((d+t)^2/n)$ and scaling displacement $O((d+t)/n)$. To compare the full Gaussian factors, set $Y = (S-I)/(n-1)$ and use skew-symmetry to obtain

$$Y^{\mathsf{T}}Y = \frac{S^{\mathsf{T}}S + I}{(n-1)^2}, \qquad Y^{\mathsf{T}}Y - \frac{S^{\mathsf{T}}S}{n^2} \succeq 0.$$

The last positive-semidefinite difference has trace $O(n^{-1})$, since $\mathrm{tr}(S^{\mathsf{T}}S) = n(n-1)$. Integrating the log-determinant derivative across this Gram difference, with a fixed spectral gap, gives $|\log \mathcal{G}(Y) - \log D_n(S)| = O(n^{-1})$. Thus the $O(n^{-1})$ comparison concerns the logarithmic Gaussian factors. Combining it with Lemma 4.3, deletion, centering and scalar normalization gives

$$|\log \mathcal{G}(B_X - P_m) - \log D_n(S)| = O(\eta).$$

The restoration identity is now per $A[R,T] = [m(n-1)/(2n)]^m e^{-\theta_X}$ per B_X. Insert Theorem 3.1 and $(1-1/n)^m = e^{-1}\exp(O((t+1)/n))$. The normalized ratio is $e^u(1+v)$ with $|u| \le K\eta$, $|v| \le K/n \le K\eta$. Choose a fixed sufficiently small δ and use $|e^u - 1| \le e^{|u|}|u|$ to obtain the claimed finite, uniform relative-error bound. □

5. Path convolution and a determinant representation

5.1 Positive path convolution

Lemma 5.1 (Path convolution).

For every tournament,

$$H(T) = \sum_{U \subseteq V(T)} \det(I + A[U]) \operatorname{per} A[U^c]. \tag{5.1}$$

Proof.

The permanent of the adjacency matrix counts directed cycle covers. The convolution equation (5.1) converts cycle-cover counts on complementary vertex sets into the path count. It is the tournament specialization of Proposition 2 of Irving and Omar [1], where the complementary adjacency matrix is $\overline{A} = J - A = I + A^{\mathsf{T}}$ and includes the diagonal entries.

The identity can also be checked directly using generating functions. Set $X = \mathrm{diag}(z_1, \dots, z_n)$, the diagonal matrix with diagonal entries $z_1, \dots, z_n$ and zero off-diagonal entries. More generally, $\mathrm{diag}(v)$ has the entries of the vector v on its diagonal. Multiplying by the corresponding variable whenever a vertex is visited gives the generating function for all walks:

$$1 + \mathbf{1}^{\mathsf{T}}(I - XA)^{-1}X\mathbf{1} = \frac{\det(I + X\overline{A})}{\det(I - XA)}.$$

The equality follows from the rank-one determinant lemma. Extracting the coefficient of $z_1 \cdots z_n$ selects exactly the Hamiltonian paths. The trace of a matrix is the sum of its diagonal entries,

$$\operatorname{tr} M = \sum_i M_{ii}.$$

For the denominator, the formal identity $\det(I - XA)^{-1} = \exp(\sum_{r\geq 1} \operatorname{tr}((XA)^r)/r)$ shows that the coefficients of squarefree monomials, in which each variable has exponent zero or one, count disjoint directed cycle covers and thus equal permanents. Combining this with the principal-minor expansion of the numerator gives

$$H(T) = \sum_U \det \overline{A}[U] \operatorname{per} A[U^c].$$

Substituting $\overline{A}[U] = I + A[U]^{\mathsf{T}}$ proves the identity.

Write $T[U]$ for the subtournament induced by the vertex set U, and let $k = |U|$. The symmetric part of $I + A[U]$ is the positive definite matrix $(I + J)/2$. Thus its real eigenvalues are positive, its nonreal eigenvalues occur in conjugate pairs, and its determinant is positive. The squared row norms are $1 + d_i^+(T[U])$, whose mean is $(k + 1)/2$. Hadamard's inequality and the arithmetic–geometric mean inequality give

$$0 < \det(I + A[U]) \leq h_k, \qquad h_k = \left(\frac{k+1}{2}\right)^{k/2}.$$

This proves the convolution and positivity claims. □

Set $h_0 = 1$ and define

$$w_k = \frac{2^k h_k}{k!}.$$

Lemma 5.1a (Fixed moments and a uniform tail). For every fixed $c > 0$ and integer $r \geq 0$,

$$\sum_{k\geq 0} (k+1)^r c^k w_k < \infty.$$

For sufficiently large n, put $k_n = \lceil 4 \log n / \log\log n \rceil$. There is an absolute constant C such that the normalized contribution of $|U| > k_n$ to (5.1) is bounded uniformly by

$$B_n := \frac{C}{2}\sqrt{n+1} \sum_{k>k_n} w_k = n^{-3/2+o(1)} = o(n^{-1}).$$

Proof. Stirling's formula gives $\log(c^k w_k) = -(k/2)\log k + O_c(k)$, and the exact ratio of successive terms is at most $K_c/\sqrt{k+1}$. Hence every fixed moment converges and $\sum_{k>k_n} c^k w_k = n^{-2+o(1)}$. Brégman's bound and degree balancing [2, Lemma 2.1 and Corollary 2.3] give $\operatorname{per} A_T \leq C\sqrt{m+1}\, m!/2^m$ for every tournament of order m; enlarge C to include $m = 0$. Consequently, the normalized convolution terms of size k are at most

$$\frac{\binom{n}{k} h_k C\sqrt{n-k+1}\,(n-k)!\,2^{-(n-k)}}{n!\,2^{-(n-1)}} \leq \frac{C}{2}\sqrt{n+1}\, w_k.$$

Summing proves the assertion. □

We call $|U| \leq k_n$ the short terms. In the error estimates below, retaining the fixed moments of $c^k w_k$ instead of bounding k by k_n avoids logarithmic losses.

5.2 The emergence of the spectral factor

First suppose $d = o(\sqrt{n})$. By Lemma 4.5, for $|U| = k \le k_n$ we have

$$\operatorname{per} A[U^c] = e^{-1} D_n(S) \frac{(n-k)!}{2^{n-k}} \left(1 + O\left(\frac{(d+k+1)^2}{n}\right)\right).$$

For integers $0 \le k \le n$, define the falling factorial by

$$(n)_k = \prod_{h=0}^{k-1} (n-h),$$

Thus $(n)_0 = 1$, and $(n)_k > 0$ in this range. The normalization satisfies $((n-k)!/2^{n-k})/\mu_n = 2^{k-1}/(n)_k$. Substituting the leading term into Lemma 5.1 therefore gives the normalized sum

$$\frac{e^{-1} D_n(S)}{2} \sum_{|U| \le k_n} \frac{2^{|U|}}{(n)_{|U|}} \det(I + A[U]),$$

The relative error is uniform over tournaments and sets $|U| \le k_n$. Replacing $(n)_k$ by n^k in the short sum and adding the tail yields the principal-minor identity below; Section 8.1 estimates the resulting errors.

$$\sum_U (2/n)^{|U|} \det(I + A[U]) = \det\left(I + \frac{2}{n}(I + A)\right).$$

Since $2A = J - I + S$, the rank-one term J asymptotically contributes a factor of two, while the scalar term contributes e. More precisely, the small-score assumption gives

$$\det\left(I + \frac{2}{n}(I + A)\right) = 2e \det(I + S/n)\left(1 + O(n^{-1} + d^2/n^2)\right).$$

The factor e^{-1} in the permanent estimate comes from restoring $(1 - 1/n)^{n-k}$ and reflects the zero diagonal. The factor $1/2$ comes from the path normalization above. They cancel the factor $2e$ in the generating determinant, leaving $D_n(S) \det(I + S/n) = \rho_n(S)$. The two determinants have separate origins: the first approximates cycle-cover counts, and the second sums the weights in the path convolution. Theorem 8.1 will complete the error summation and prove

$$\frac{H(T)}{\mu_n} = \rho_n(S) + O((d+1)^2/n) \qquad \text{when } d = o(\sqrt{n}).$$

To obtain an upper bound for all tournaments, Section 7 separately handles large score variance and extreme degrees, retaining the score penalty in the remaining class. The next lemma controls this same spectral factor for every tournament.

6. The spectral bound and equality

The spectral-ratio bound in this section is due to Divoux and Wang [12, Proposition 7.1], using the comparison of Fang and Huang [11]. The determinant polynomial and the carousel switching are also in [12, Section 7 and Lemma 9.1]. We give the details and follow the equality conditions needed for the fixed-base corollary.

6.1 The spectral radius and an elementary uniform bound

Lemma 6.1 (Spectral cap). Every tournament sign matrix of order n satisfies

$$\|S\|_{\mathrm{op}} \le \cot\left(\frac{\pi}{2n}\right) < \frac{2n}{\pi}, \qquad \rho_n(S) \le \frac{3n-1}{n+1} < 3.$$

Proof. The phase-ordering argument [4, Theorem 3.1] gives the operator bound as follows. Apply a simultaneous signed permutation to z and S so that the phases of the nonzero entries of z lie in increasing order in $[0,\pi)$. Then $c_{ij} = \mathrm{Im}(\overline{z}_i z_j) \ge 0$ for $i<j$, and, for the transitive sign matrix T_n^0,

$$z^* i S z = -2\sum_{i<j} S_{ij} c_{ij} \le 2\sum_{i<j} c_{ij} = z^*(-iT_n^0)z.$$

Rayleigh's principle reduces the bound to T_n^0. Subtracting consecutive row eigenvalue equations gives $(\lambda-1)v_i = (\lambda+1)v_{i+1}$. The geometric ratio $r = (\lambda-1)/(\lambda+1)$ satisfies $r^n = -1$ by the first row equation. Thus the eigenvalues are $i\cot((2j-1)\pi/(2n))$, $1 \le j \le n$. Their largest modulus is $\cot(\pi/(2n))$, and $\tan u > u$ gives the strict inequality (the case $n=1$ is immediate).

For the ratio bound, write the eigenvalues in nonzero pairs $\pm i\lambda_j$ and put $x_j = \lambda_j^2/n^2$. The Frobenius identity gives $\sum_j x_j = (n-1)/(2n) < 1/2$. With $\phi(x) = \log((1+x)/(1-x))$, the derivative $\phi'(x) = 2/(1-x^2)$ is increasing on $[0,1)$ and $\phi(0)=0$. Hence $\phi(a+b) \ge \phi(a)+\phi(b)$ for $a,b \ge 0$, $a+b<1$. Repeatedly merging the x_j gives

$$\log \rho_n(S) = \sum_j \phi(x_j) \le \phi\left(\frac{n-1}{2n}\right),$$

which proves the stated bound, including $n=1$.

The eigenvalue calculation also yields

$$\det(I + zT_n^0) = \frac{(1+z)^n + (1-z)^n}{2}.$$

Indeed, each even transitive principal determinant equals one: the positive cotangent frequencies pair with their reciprocals at complementary angles. Odd principal determinants vanish by skew-symmetry. The principal-minor expansion is therefore $\sum_h \binom{n}{2h} z^{2h}$. □

Historical remark. Combining the stronger cap $x_j \le 4/\pi^2$ with $\sum_j x_j \le 1/2$ gives the earlier constant

$$C_* = \frac{3\pi^4 + 4\pi^2 - 32}{\pi^4 + 4\pi^2 - 32} > L.$$

It is the relaxed convex maximum attained at $4/\pi^2, 1/2 - 4/\pi^2, 0, \ldots$. The present proof uses the elementary bound $\rho_n < 3$ to exclude exceptional vertices and the all-order sharp comparison in Proposition 6.6 for the final upper bound.

6.2 Product comparison and equality

Theorem 6.2 (Fang–Huang).

We use the following form of Fang–Huang, Corollary 3.4. Define the comparison matrix K_d by

$$(K_d)_{ij} = \begin{cases} 1, & i+j \le d, \\ 0, & i+j = d+1, \\ (-1)^{i+j-d-2}, & i+j \ge d+2. \end{cases}$$

Let $p \ge 1$, let $x \in \mathbb{R}^d$, and let $S_1, \ldots, S_p$ be tournament sign matrices of order d. Write x^* for the vector obtained by arranging the absolute values of the coordinates of x in decreasing order. Then

$$(S_1 \cdots S_p x)^* \prec_w K_d^p x^*.$$

The right-hand side is a nonnegative decreasing vector. Weak majorization $u \prec_w v$ means

$$\sum_{i=1}^{r} u_i \le \sum_{i=1}^{r} v_i \quad (1 \le r \le d).$$

Moreover, vector equality in the outermost comparison implies equality for every suffix product. To apply this to Frobenius norms, we first record the squared-norm inequality under weak majorization, including its equality case.

Lemma 6.3 (Truncated sums under weak majorization).

Write $(a-t)_+ = \max(a-t, 0)$. For nonnegative decreasing vectors satisfying $u \prec_w v$ and $t \ge 0$,

$$\sum_{i=1}^{d} (u_i - t)_+ = \max_{0 \le r \le d} \left(\sum_{i=1}^{r} u_i - rt \right) \le \max_{0 \le r \le d} \left(\sum_{i=1}^{r} v_i - rt \right) = \sum_{i=1}^{d} (v_i - t)_+.$$

Proof.

The first equality follows by taking $r = \#\{i : u_i > t\}$: adding coordinates at most t, or removing coordinates greater than t, cannot increase the sum. For $a \ge 0$, we also have

$$2 \int_0^\infty (a-t)_+ \, dt = 2 \int_0^a (a-t) \, dt = 2[at - t^2/2]_0^a = a^2.$$

Integrating coordinate by coordinate gives

$$\|u\|_2^2 = 2 \int_0^\infty \sum_i (u_i - t)_+ \, dt \le 2 \int_0^\infty \sum_i (v_i - t)_+ \, dt = \|v\|_2^2.$$

Suppose the squared norms are equal. The function $g(t) = \sum_i (v_i - t)_+ - \sum_i (u_i - t)_+$ is continuous and nonnegative, with integral zero. If $g(t_0) > 0$, then $g(t) > g(t_0)/2$ on an interval of positive length, a contradiction. Thus $g \equiv 0$. Comparing right derivatives gives

$$\#\{i : u_i > t\} = \#\{i : v_i > t\} =: N(t).$$

For coordinate j,

$$u_j = \int_0^\infty \mathbf{1}_{\{\#\{i : u_i > t\} \ge j\}} \, dt = \int_0^\infty \mathbf{1}_{\{N(t) \ge j\}} \, dt = v_j.$$

Equality of squared norms therefore forces $u = v$. Taking $x = e_j$ and using $e_j^* = e_1$ yields

$$\|S^p e_j\|_2^2 \le \|K_d^p e_1\|_2^2.$$

Equality also forces $(S^p e_j)^* = K_d^p e_1$.

6.3 Moments of the comparison matrix

Write $\operatorname{tr} M = \sum_j M_{jj}$.

Let T_d be the transitive tournament sign matrix, with all entries above the diagonal equal to 1. Let R_0 be the reversal permutation matrix and $D_0 = \operatorname{diag}(1, -1, \ldots)$. By the definition of K_d,

$$(D_0K_dR_0)_{ij} = (-1)^{i-1}(K_d)_{i,d+1-j} = \begin{cases} (-1)^{i-1}, & i < j, \\ 0, & i = j, \\ (-1)^j, & i > j. \end{cases}$$

Here $R_0e_j = e_{d+1-j}$ and $(D_0)_{jj} = (-1)^{j-1}$. Since $(K_d)_{ij}$ depends only on $i + j$, we have $K_d^T = K_d$. Write the sequence of increasing odd indices followed by decreasing even indices as

$$(\ell_1, \dots, \ell_d) = (1, 3, \dots, d, d-1, d-3, \dots, 2), \qquad \Lambda e_j = e_{\ell_j}.$$

Each index occurs once. Hence $\Lambda^T\Lambda = \Lambda\Lambda^T = I$, and $\ell_1 = 1$.

Permute the rows and columns in this order. Odd indices are increasing, so the corresponding upper-triangular entries are $(-1)^{i-1} = 1$; even indices are decreasing, so those entries are $(-1)^j = 1$. An odd index precedes an even index, and the corresponding entry is $(-1)^{i-1}$ if $i < j$, or $(-1)^j$ if $i > j$, again equal to 1. Hence

$$T_d = \Lambda^T D_0 K_d R_0 \Lambda.$$

Set $U = D_0\Lambda$. Since $R_0R_0^T = I$, $\Lambda\Lambda^T = I$ and $K_d^T = K_d$, multiplication gives

$$\begin{aligned} T_dT_d^T &= \Lambda^T D_0 K_d R_0 \Lambda\Lambda^T R_0^T K_d D_0 \Lambda \\ &= \Lambda^T D_0 K_d^2 D_0 \Lambda = U^T K_d^2 U. \end{aligned}$$

The first index in this ordering is 1, so $Ue_1 = e_1$. Also $T_d^T = -T_d$, hence

$$T_d^2 = -U^T K_d^2 U, \qquad T_d^{2p} = (-1)^p U^T K_d^{2p} U$$

because each adjacent pair UU^T in the product equals I. Consequently,

$$(-1)^p (T_d^{2p})_{11} = e_1^T K_d^{2p} e_1 = (K_d^p e_1)^T (K_d^p e_1) = \|K_d^p e_1\|_2^2.$$

The signed cyclic shift Z in the proof companion, Appendix A.2 satisfies $Z^T T_d Z = T_d$, and therefore $Z^T T_d^{2p} Z = T_d^{2p}$. Comparing diagonal entries gives $(T_d^{2p})_{j+1,j+1} = (T_d^{2p})_{jj}$ for $j < d$. At $j = d$, the two signs cancel and give $(T_d^{2p})_{11} = (T_d^{2p})_{dd}$. Thus all diagonal entries agree, and

$$\begin{aligned} \|T_d^p\|_F^2 &= \operatorname{tr}((T_d^p)^T T_d^p) = (-1)^p \operatorname{tr} T_d^{2p} \\ &= d(-1)^p (T_d^{2p})_{11} = d\|K_d^p e_1\|_2^2. \end{aligned}$$

Sum the d column inequalities from the preceding section:

$$\|S^p\|_F^2 = \sum_{j=1}^{d} \|S^p e_j\|_2^2 \le d\|K_d^p e_1\|_2^2 = \|T_d^p\|_F^2 \qquad (p \ge 1).$$

In particular, the inequality holds for every odd p, as required in the next section.

6.4 The extremum of the spectral ratio

Let S be any tournament sign matrix of order d, with nonzero eigenvalues $\pm i\lambda_j$, where $\lambda_j > 0$, and put $x_j = \lambda_j^2/d^2$. Since $2\sum_j \lambda_j^2 = \|S\|_F^2 = d(d-1)$, we have $x_j \le (d-1)/(2d) < 1/2$. In the skew-symmetric canonical form, a zero block contributes 1 to both determinants, while a nonzero two-dimensional block contributes $1 + x_j$ and $1 - x_j$, respectively. Hence

$$\det(I + S/d) = \prod_j (1 + x_j), \qquad \det(I + iS/d) = \prod_j (1 - x_j) > 0.$$

For $0 \le x < 1$, integrate the geometric series:

$$\begin{aligned}\log \frac{1+x}{1-x} &= \int_0^x \left(\frac{1}{1+t} + \frac{1}{1-t} \right) dt \\ &= 2 \int_0^x \frac{dt}{1-t^2} = 2 \sum_{r=0}^{\infty} \int_0^x t^{2r}\, dt = 2 \sum_{r=0}^{\infty} \frac{x^{2r+1}}{2r+1}.\end{aligned}$$

The interchange follows from uniform convergence of the geometric series on $[0, x]$. The skew-symmetric canonical form also gives $\|S^p\|_F^2 = 2 \sum_j \lambda_j^{2p}$. Substituting term by term yields

$$\begin{aligned}\log \rho_d(S) &= 2 \sum_j \sum_{r=0}^{\infty} \frac{\lambda_j^{4r+2}}{(2r+1) d^{4r+2}} \\ &= \sum_{r=0}^{\infty} \frac{\|S^{2r+1}\|_F^2}{(2r+1) d^{4r+2}} \le \sum_{r=0}^{\infty} \frac{\|T_d^{2r+1}\|_F^2}{(2r+1) d^{4r+2}} = \log \rho_d(T_d).\end{aligned}$$

Put $r_d = \rho_d(T_d)$. If equality holds above, every nonnegative difference in the series vanishes. The term $r = 1$ gives $\|S^3\|_F^2 = \|T_d^3\|_F^2$. Since this difference is itself the sum of the nonnegative column differences, equality holds in each column. The equality criterion of Section 6.2 and the suffix equality in Fang–Huang's theorem then give

$$(S^3 e_j)^* = K_d^3 e_1, \qquad (S^2 e_j)^* = K_d^2 e_1.$$

Take squared norms in the second identity and sum over j:

$$\operatorname{tr} S^4 = \|S^2\|_F^2 = d \|K_d^2 e_1\|_2^2 = \|T_d^2\|_F^2 = \operatorname{tr} T_d^4.$$

By the proof companion, Appendix A.3, equality of the fourth moments is equivalent to $S = \Omega^T T_d \Omega$ for a signed permutation matrix Ω. Conversely, $\Omega^T \Omega = I$ implies

$$I + tS = \Omega^T (I + tT_d) \Omega, \qquad \det(I + tS) = (\det \Omega)^2 \det(I + tT_d) = \det(I + tT_d).$$

Lemma 6.4 (The spectral ratio and equality).

Take $t = 1/d$ and $t = i/d$ separately to obtain $\rho_d(S) = r_d$. Thus

$$\rho_d(S) \le r_d, \quad \text{with equality if and only if } S = \Omega^T T_d \Omega.$$

Proof (Ordinary isomorphism in the regular case).

Write $\Omega = D_\varepsilon P$, where P is a permutation matrix and $D_\varepsilon = \operatorname{diag}(\varepsilon_1, \dots, \varepsilon_d)$ with $\varepsilon_i \in \{-1, 1\}$. Put $\varepsilon = (\varepsilon_1, \dots, \varepsilon_d)^T$. Then

$$S = P^T D_\varepsilon T_d D_\varepsilon P, \qquad P S P^T = D_\varepsilon T_d D_\varepsilon.$$

Relabel by P and continue to write S for the resulting matrix. If the tournament is regular, then $S\mathbf{1} = 0$, and $D_\varepsilon^2 = I$ gives

$$D_\varepsilon T_d \varepsilon = 0, \qquad T_d \varepsilon = 0.$$

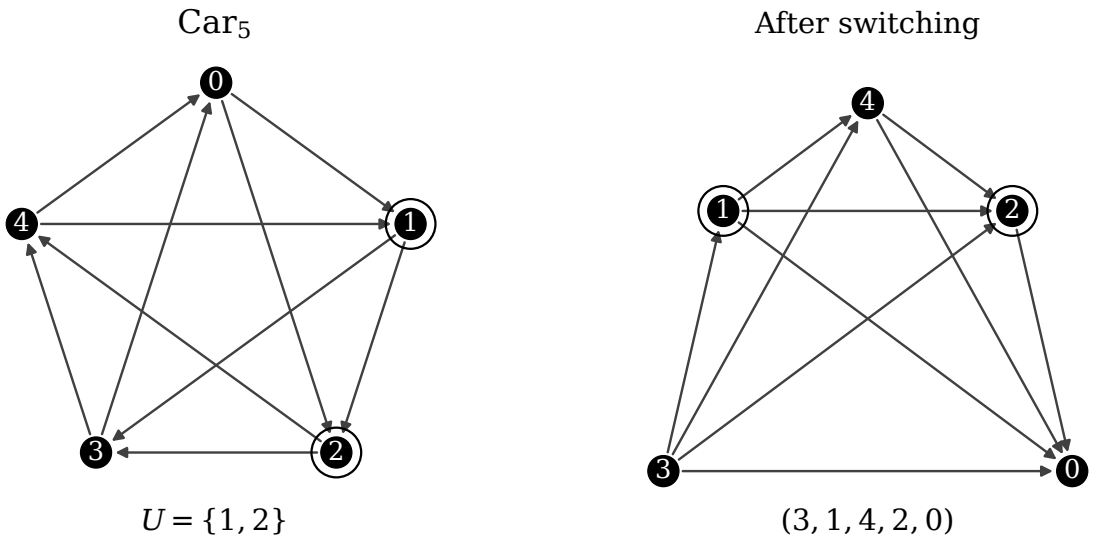


Figure 3: Reversing all arcs between $U = \{1, 2\}$ and its complement in Car_5 gives the transitive order $(3, 1, 4, 2, 0)$. Circled vertices belong to U. This switching conjugates the sign matrix as $S \mapsto DSD$ and preserves the spectral ratio; it need not preserve the Hamiltonian path count.

The difference between rows i and $i + 1$ has entry 1 in each of columns $i, i + 1$ and zero elsewhere. Therefore

$$0 = (T_d\varepsilon)_i - (T_d\varepsilon)_{i+1} = \varepsilon_i + \varepsilon_{i+1}.$$

It follows that $\varepsilon_i = (-1)^{i-1}\varepsilon_1$. The common sign ε_1 cancels in $D_\varepsilon T_d D_\varepsilon$, leaving the alternating switching. the proof companion, Appendix A.2 identifies this tournament with Car_d. Thus, among regular tournaments, equality holds only for the carousel, up to isomorphism.

6.5 A gap from the sixth moment

Lemma 6.5 (Strict sixth-moment deficit).

If regular S is not ordinarily isomorphic to the carousel, the preceding equality argument gives

$$\Delta_3 := \|T_d^3\|_F^2 - \|S^3\|_F^2 > 0.$$

Proof.

Modulo 2, $S \equiv J + I$. Since $J^2 = dJ$ and $J^3 = d^2J$,

$$S^3 \equiv (J + I)^3 = I + 3J + 3dJ + d^2J \equiv I + (1 + d + d^2)J \equiv I + J \pmod 2.$$

The last step uses that $d(d + 1)$ is even. Also $(S^3)^T = -S^3$, so its diagonal entries vanish and its off-diagonal entries are odd. If $z = 2k + 1$,

$$z^2 = 4k(k + 1) + 1 \equiv 1 \pmod 8.$$

Pair the skew-symmetric positions in the squared norm:

$$\|S^3\|_F^2 = 2\sum_{i<j}((S^3)_{ij})^2 \equiv 2\binom{d}{2} = d(d - 1) \pmod{16}.$$

This also applies to T_d. Thus Δ_3 is a positive multiple of 16, and $\Delta_3 \geq 16$. Retaining just term $r = 1$ in the logarithmic series gives

$$\log \frac{r_d}{\rho_d(S)} \geq \frac{\Delta_3}{3d^6} \geq \frac{16}{3d^6}.$$

Exponentiate and rearrange:

$$\rho_d(S) \leq r_d e^{-16/(3d^6)}, \qquad r_d - \rho_d(S) \geq r_d(1 - e^{-16/(3d^6)}) > 0.$$

The gap depends on the base order d. Section 11.2 obtains a corresponding block threshold by comparing the limits.

6.6 The sharp comparison at every order

Proposition 6.6 (All-order spectral comparison). For every tournament sign matrix S of order n, $\rho_n(S) \leq r_n$, where $r_n = L + O(n^{-1})$ is the explicit quotient below. The $O(n^{-1})$ constant is absolute and uniform over tournaments.

Proof.

For $n = 1$ the ratio equals one. The regular equality classification in Section 6 concerns odd orders, but the preceding spectral inequality does not require regularity. To make its use at every order explicit, let $n \geq 2$ and write the permutation in Section 6.3 as all odd indices in increasing order followed by all even indices in decreasing order. At odd order this is the original permutation. At even order it is $(1, 3, \ldots, n-1, n, n-2, \ldots, 2)$. The same entrywise computation gives

$$T_n = \Lambda^T D_0 K_n R_0 \Lambda, \qquad T_n^2 = -U^T K_n^2 U, \qquad U = D_0 \Lambda, \qquad U e_1 = e_1.$$

The signed wrap-around shift satisfies $Z^T T_n Z = T_n$ at either parity. Consequently, all diagonal entries of T_n^{2p} coincide. For every integer $p \geq 1$, the columnwise Fang–Huang comparison and these identities give

$$\|S^p\|_F^2 \leq n \|K_n^p e_1\|_2^2 = \|T_n^p\|_F^2.$$

Substitution into the absolutely convergent logarithmic series of Section 6.4 yields, at every order,

$$\log \rho_n(S) = \sum_{r \geq 0} \frac{\|S^{2r+1}\|_F^2}{(2r+1)n^{4r+2}} \leq \log \rho_n(T_n).$$

The determinant polynomial in Lemma 6.1 also holds at both parities. Hence

$$\rho_n(S) \leq r_n = \frac{(1 + 1/n)^n + (1 - 1/n)^n}{(1 + i/n)^n + (1 - i/n)^n} = L + O(n^{-1}).$$

For the last step, the Taylor estimate for the logarithm gives, for $z \in \{1, -1, i, -i\}$, $n \log(1 + z/n) = z + O(n^{-1})$, while the denominator tends to $2 \cos 1 > 0$. The quotient therefore has an $O(n^{-1})$ error. These estimates depend on the order alone, not on the tournament. □

7. The upper bound for arbitrary tournaments

Write $V_{\text{deg}}(T)$ for degree variance, reserving $V(T)$ for the vertex set:

$$V_{\text{deg}}(T) = \sum_i \left(d_i^+ - \frac{n-1}{2} \right)^2 = \frac{1}{4} \|S\mathbf{1}\|_2^2, \qquad \tau = \frac{4V_{\text{deg}}(T)}{(n-1)^2}.$$

The proof has three cases. If $V_{\deg}(T) \geq 16n^2 \log n$, the variance penalty in Brégman's inequality makes the path count negligible. Otherwise, vertices with degrees outside $[(n-1)/20, 19(n-1)/20]$ number at most $O(\log n)$. If any such vertices exist, separating them from the rest gives a path count below the random expectation. In the remaining tournaments, normalized scores stay a fixed distance from ± 1, and $\tau = O(\log n)$. For them, the permanent estimate below retains the factor $e^{-\tau}$ and controls the final error by $O(n^{-1})$.

The constant K in this section may increase from one occurrence to the next; after the degree thresholds above are fixed, it is absolute. Section 5.1 has already bounded the long terms by $o(n^{-1})\mu_n$, so here we estimate only short terms $|U| \leq k_n$.

The three cases require different precision: the high-variance class contributes $o(n^{-1})$ after normalization; the exceptional-vertex class only needs a fixed gap below the baseline 1; the remaining class carries the sharp $O(n^{-1})$ estimate. All constants in the local scaling lemmas are fixed before taking the common large-order threshold.

Proposition 7.1 (Uniform sharp upper bound). There are absolute constants K, n_0 such that every tournament T of order $n \geq n_0$ satisfies

$$H(T) \leq (L + K/n)\mu_n.$$

7.1 Applying the nonprincipal permanent estimate

The estimate of Lemma 4.4 allows different deleted row and column sets. We use it below to control edges between the exceptional set and the core. When the deleted sets agree, the score product is retained outside the path convolution.

7.2 Large score variance

Lemma 7.2 (Permanent bound with a variance penalty).

There is an absolute constant K such that, for every order $m \geq 1$, a tournament Q with outdegrees d_i and degree variance $V_{\deg}(Q) = \sum_i (d_i - (m-1)/2)^2$ satisfies

$$\operatorname{per} A_Q \leq K\sqrt{m+1}\frac{m!}{2^m} \exp\left(-\frac{V_{\deg}(Q)}{8m^2}\right).$$

Proof.

Concavity of the row-degree factor strengthens the Brégman bound [2]. If any $d_i = 0$, the bound is immediate. Otherwise, define $f(k) = \log(k!)/k$ for positive integers. Direct calculation gives, for $k \geq 2$,

$$2f(k) - f(k-1) - f(k+1) = \frac{2[\log k - f(k-1)] - k\log(1+1/k)}{k(k+1)}.$$

Applying the arithmetic–geometric mean inequality to $1, \ldots, k-1$ gives $\log k - f(k-1) \geq \log 2$. Also, $(1+1/k)^k < e < 3$, so the numerator is at least $\log(4/3) \geq 1/4$. Consequently, on the degree interval $[1, m-1]$, the piecewise linear interpolation of the integer-point function $f(k) + k^2/(8m^2)$ is concave. Jensen's inequality at the mean $\eta = (m-1)/2$ gives

$$\sum_i f(d_i) \leq m\widetilde{f}(\eta) - \frac{V_{\deg}(Q)}{8m^2} + \frac{1}{32m},$$

Here $\widetilde{f}$ is the linear interpolation. The last term handles a half-integer mean and can be omitted when the mean is an integer. Stirling's formula gives

$$\exp(m\widetilde{f}(\eta)) \leq K\sqrt{m+1}\frac{m!}{2^m}.$$

Combining this with Brégman's row-degree bound proves the variance estimate.

We apply this bound to the tournament associated with each short deleted set. We must therefore check that its variance remains sufficiently large after deletion.

After deleting a size-k set U, each remaining centered degree changes by at most $k/2$, and the sum of the original squared deviations at deleted vertices is at most $kn^2/4$. Applying Cauchy–Schwarz to the cross term gives

$$V_{\deg}(T-U) \geq V_{\deg}(T) - \frac{kn^2}{4} - k\sqrt{nV_{\deg}(T)}.$$

If $V_{\deg}(T) \geq 16n^2 \log n$, then uniformly for every $k \leq k_n$,

$$V_{\deg}(T-U) \geq (1-o(1))V_{\deg}(T).$$

Thus the variance penalty in every short permanent is at most $n^{-2+o(1)}$, absorbing the preceding $\sqrt{n}$ factor. Summation with the summable determinant weights bounds the total normalized short contribution by $n^{-3/2+o(1)}$. Adding the long-deletion tail gives, uniformly over this graph class,

$$V_{\deg}(T) \geq 16n^2 \log n \quad \Longrightarrow \quad H(T)/\mu_n = o(1/n).$$

Write $L_k = \log(k!)$. Then

$$\begin{aligned} 2f(k) - f(k-1) - f(k+1) &= \frac{2(L_{k-1} + \log k)}{k} - \frac{L_{k-1}}{k-1} - \frac{L_{k-1} + \log k + \log(k+1)}{k+1} \\ &= \frac{2\log k - 2L_{k-1}/(k-1) - k\log(1+1/k)}{k(k+1)}. \end{aligned}$$

AM–GM gives

$$((k-1)!)^{1/(k-1)} \leq \frac{1 + \cdots + (k-1)}{k-1} = k/2,$$

Hence $\log k - f(k-1) \geq \log 2$. Also, $k\log(1+1/k) < 1 < \log 3$, and

$$\log(4/3) = \int_1^{4/3} \frac{du}{u} \geq \frac{1/3}{4/3} = \frac{1}{4}.$$

The second difference is therefore at least $1/[4k(k+1)] \geq 1/(4m^2)$. Set $h(k) = f(k) + k^2/(8m^2)$. Then

$$2h(k) - h(k-1) - h(k+1) \geq \frac{1}{4m^2} - \frac{2}{8m^2} = 0.$$

If η is an integer, $\widetilde{h}(\eta) = \widetilde{f}(\eta) + \eta^2/(8m^2)$. If $\eta = k + 1/2$, then

$$\frac{k^2 + (k+1)^2}{2} = \eta^2 + \frac{1}{4}, \quad \widetilde{h}(\eta) = \widetilde{f}(\eta) + \frac{\eta^2 + 1/4}{8m^2}.$$

Jensen's inequality therefore gives

$$\begin{aligned} \sum_i f(d_i) + \frac{\sum_i d_i^2}{8m^2} = \sum_i h(d_i) &\leq m\widetilde{h}(\eta) \\ &\leq m\widetilde{f}(\eta) + \frac{m\eta^2}{8m^2} + \frac{1}{32m}, \\ \sum_i f(d_i) &\leq m\widetilde{f}(\eta) - \frac{\sum_i (d_i - \eta)^2}{8m^2} + \frac{1}{32m}. \end{aligned}$$

For $m = 2r + 1$, the logarithmic form of Stirling's formula gives

$$mf(r) = \frac{2r+1}{r}\left[r\log r - r + \tfrac{1}{2}\log(2\pi r) + O(1/r)\right],$$

$$\log(m!/2^m) = m\log(m/2) - m + \tfrac{1}{2}\log(2\pi m) + O(1/m).$$

Subtracting and using $r = (m-1)/2$ gives $\frac{1}{2}\log m + O(1)$. For $m = 2r$, $\widetilde{f}(\eta) = [f(r-1) + f(r)]/2$; applying the same formula to each term gives the same difference. Exponentiating therefore yields $K\sqrt{m+1}$. If $d_i = 0$, each term in the permanent contains a zero factor. The cases $m = 1, 2$ are incorporated directly into the constant.

Write $b_i = d_i^+ - (n-1)/2$. For a retained vertex, let e_i count its outgoing edges to U. Then

$$b_i' = d_i^+ - e_i - (n-k-1)/2 = b_i + \delta_i, \qquad \delta_i = k/2 - e_i, \quad |\delta_i| \le k/2.$$

Thus

$$\begin{aligned} V_{\deg}(T-U) &= \sum_{i\notin U}(b_i+\delta_i)^2 \\ &\ge V_{\deg}(T) - \sum_{i\in U} b_i^2 - 2\left|\sum_{i\notin U} b_i\delta_i\right| \\ &\ge V_{\deg}(T) - kn^2/4 - k\sum_{i\notin U}|b_i| \\ &\ge V_{\deg}(T) - kn^2/4 - k\sqrt{nV_{\deg}(T)}. \end{aligned}$$

When $V \ge 16n^2\log n$, the relative losses are respectively at most

$$\frac{k}{64\log n} = o(1), \qquad k\sqrt{n/V} \le \frac{k}{4\sqrt{n\log n}} = o(1).$$

It follows that $V_{\deg}(T-U)/(8(n-k)^2) \ge (2-o(1))\log n$, so the penalty is $n^{-2+o(1)}$. Multiplying by $\sqrt{n}$ and $\sum_k w_k < \infty$ gives $n^{-3/2+o(1)}$.

To estimate the factorial factor in the variance bound, apply the Stirling formula from the proof companion, Appendix B.3. With c denoting its constant,

$$f(r) = \log r - 1 + \frac{\log r}{2r} + \frac{c}{r} + O(r^{-2}).$$

First take $m = 2r + 1$, so that the average outdegree is the integer r. Multiplication by m gives

$$mf(r) = m\log r - m + \left(1 + \frac{1}{2r}\right)\log r + 2c + \frac{c}{r} + O(r^{-1}).$$

The logarithm of the natural scale is

$$\log(m!/2^m) = m\log(m/2) - m + \frac{1}{2}\log m + c + O(m^{-1}).$$

Denote their difference by Δ_m:

$$\begin{aligned} \Delta_m &= m\log\frac{2r}{m} + \log r - \frac{1}{2}\log m + c + O(\log m/m) \\ &= m\log(1-1/m) + \frac{1}{2}\log m - \log 2 + \log(1-1/m) + c + O(\log m/m). \end{aligned}$$

Since $m\log(1-1/m) = -1 + O(1/m)$ and all other terms except $\frac{1}{2}\log m$ are uniformly bounded, $\Delta_m \le \frac{1}{2}\log m + K$.

Next take $m = 2r$ with $r \ge 2$. The mean is $r - 1/2$, and linear interpolation gives

$$m\widetilde{f}((m-1)/2) = r[f(r-1) + f(r)].$$

Substitution term by term yields

$$\begin{aligned} r[f(r-1)+f(r)] = {} & r\log(r-1) + r\log r - 2r \\ & + \frac{r}{2(r-1)}\log(r-1) + \frac{1}{2}\log r \\ & + c\frac{r}{r-1} + c + O(r^{-1}). \end{aligned}$$

Subtracting $\log((2r)!/2^{2r}) = 2r\log r - 2r + \frac{1}{2}\log(2r) + c + O(r^{-1})$ gives

$$\begin{aligned} \Delta_{2r} = {} & r\log(1-1/r) + \frac{r}{2(r-1)}\log(r-1) \\ & - \frac{1}{2}\log 2 + c\frac{r}{r-1} + O(r^{-1}). \end{aligned}$$

The first term is nonpositive. The second satisfies

$$\frac{r}{2(r-1)}\log(r-1) \le \frac{1}{2}\log r + \frac{\log r}{2(r-1)} \le \frac{1}{2}\log r + \frac{1}{2},$$

Here $\log r \le r - 1$. The constant term $cr/(r-1)$ has absolute value at most $2|c|$. Thus again $\Delta_m \le \frac{1}{2}\log m + K$. Exponentiating in both parity cases gives

$$\exp(m\widetilde{f}((m-1)/2)) \le e^K\sqrt{m}\, m!/2^m.$$

The finitely many values of m outside the expansion range are covered by increasing the same K. □

7.3 Exceptional vertices

Assume from now on that $V_{\text{deg}}(T) < 16n^2\log n$. Define the exceptional set

$$F = \left\{ i : \frac{d_i^+}{n-1} \notin [1/20, 19/20] \right\}, \qquad f = |F|, \qquad M = T - F, \qquad N = n - f.$$

Each exceptional vertex contributes to $V_{\text{deg}}(T)$ at least a fixed positive multiple of n^2, so $f = O(\log n)$. A nonexceptional vertex originally has absolute score at most $0.9(n-1)$, and deleting F changes its score by at most f. Consequently, uniformly for all sufficiently large n,

$$\max_i |(S_M\mathbf{1})_i/(N-1)| < 0.95, \qquad \tau_M = O(\log n).$$

For the squared score sum, use $\|S_M\mathbf{1}\|_2 \le \|S\mathbf{1}\|_2 + f\sqrt{N}$ and divide by $N-1$. The same argument remains valid after further deleting a short core subset. Throughout these deletions the original exceptional set F is retained, allowing the constants to be chosen uniformly.

First classify cycle covers according to their connections between F and the core M, thereby estimating per A_T. An exceptional vertex has few neighbors in one direction. A cycle cover must use edges in both directions unless that vertex is matched within F, and the latter choice incurs a cost of order $1/N$. Use the paired core weights ℓ, r from Lemma 4.4, whose total deviation satisfies

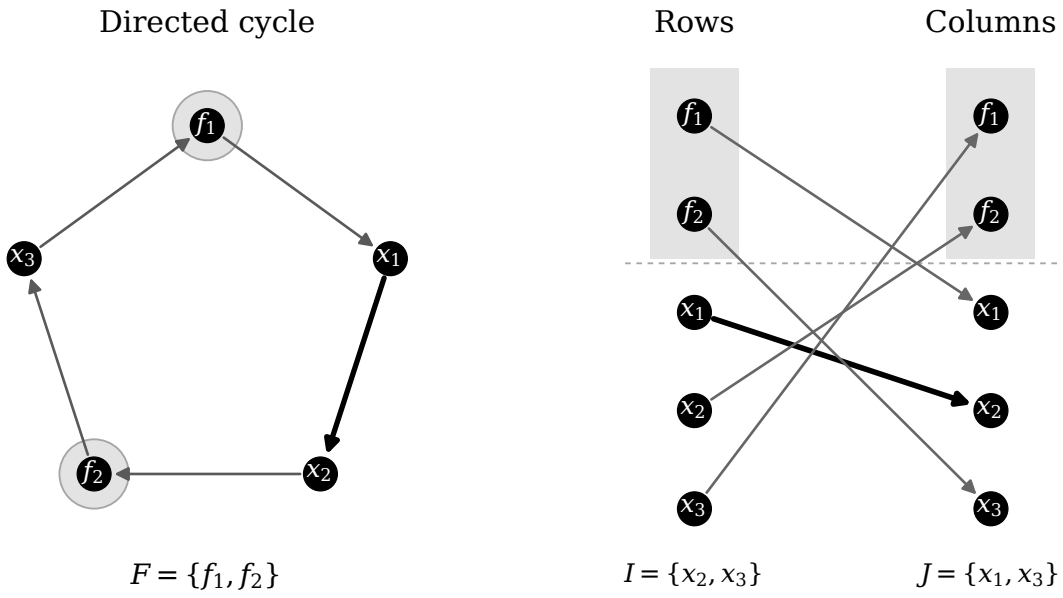


Figure 4: A directed five-cycle in a permanent term and its row-column matching. Shaded vertices form the exceptional set $F = \{f_1, f_2\}$. Cross-block arcs occupy core rows $I = \{x_2, x_3\}$ and core columns $J = \{x_1, x_3\}$, leaving the nonprincipal entry $A[x_1, x_2]$ shown in bold. Only the arcs of this permanent term are drawn.

$$\sum_{i \in M} |\ell_i - 1| + \sum_{i \in M} |r_i - 1| \le K\sqrt{N \log n}.$$

For $x \in F$, write $q_x = N^{-1}|\{j \in M : x \to j\}|$. Its distance from $[0, 1/20] \cup [19/20, 1]$ is $O(f/n)$. The weighted sums of cross-edge neighbors on the two sides are bounded by $N(q_x + \beta)$ and $N(1 - q_x + \beta)$, where $\beta = O(\sqrt{\log n/n})$. Set

$$u_x = 2(q_x + \beta), \qquad v_x = 2(1 - q_x + \beta).$$

Uniformly for every x, we may choose

$$u_x v_x \le c_n = 19/100 + o(1), \qquad u_x, v_x \le L_n = 2 + o(1).$$

Decompose the permanent according to the permutation edges lying inside F. If there are s internal edges, their row set R and column set C both have size s, and there are exactly $t = f - s$ cross edges in each direction. The deleted core row and column sets I, J both have size t, but need not coincide. The exact four-block expansion is

$$\operatorname{per} A_T = \sum_{s=0}^{f} \sum_{\substack{R,C \subseteq F \\ |R|=|C|=s}} \sum_{\substack{I,J \subseteq M \\ |I|=|J|=f-s}} \operatorname{per} A[R, C] \operatorname{per} A[F \setminus R, J] \operatorname{per} A[I, F \setminus C] \operatorname{per} A[M \setminus I, M \setminus J].$$

Every contributing permutation belongs to exactly one class. First apply the uniform bound of Lemma 4.4 to the core permanent and then separate the two cross-edge sums. Deleting a core row contributes ℓ_i, and deleting a core column contributes r_j. For fixed R, C, the outgoing cross-edge sum is exactly

$$\sum_{\substack{J\subseteq M\\|J|=t}} \operatorname{per} A[F\setminus R, J]\prod_{j\in J} r_j$$
$$= \sum_{\phi:F\setminus R\hookrightarrow M}\prod_{x\in F\setminus R} A_{x,\phi(x)}r_{\phi(x)} \le \prod_{x\in F\setminus R}\sum_{j\in M} A_{xj}r_j.$$

Independently, the incoming cross-edge sum is

$$\sum_{\substack{I\subseteq M\\|I|=t}} \operatorname{per} A[I, F\setminus C]\prod_{i\in I} \ell_i$$
$$= \sum_{\psi:F\setminus C\hookrightarrow M}\prod_{y\in F\setminus C} A_{\psi(y),y}\ell_{\psi(y)} \le \prod_{y\in F\setminus C}\sum_{i\in M} A_{iy}\ell_i.$$

Here $\hookrightarrow$ means an injection. Each injection is counted exactly once by its image set and its corresponding permanent term, with no extra factor $t!$. The image sets I, J need neither coincide nor be disjoint: they belong to the row and column copies of the core, respectively. Since every factor is nonnegative, the injectivity restrictions may be dropped. The product of the two upper bounds is

$$N^{2t}2^{-2t}\left(\prod_{x\notin R} u_x\right)\left(\prod_{y\notin C} v_y\right).$$

The permanent inside the exceptional set is at most $s!$. After division by $n!/2^n$, the full factor restoring factorials and powers of two is

$$2^s N^{2t}\frac{(N-t)!}{(N+f)!} = \left(\frac{2}{N}\right)^s \exp(O(f^2/N)).$$

Here $N+f-(N-t) = f+t = 2f-s$, so the factorial ratio contributes $N^{-(2f-s)}\exp(O(f^2/N))$. Multiplication by N^{2t} leaves exactly N^{-s}.

Let $h = |R\cap C|$. The paired product satisfies

$$\left(\prod_{x\notin R} u_x\right)\left(\prod_{y\notin C} v_y\right) \le c_n^{f-2s}L_n^{2s}(c_n/L_n^2)^h \le c_n^{f-2s}L_n^{2s}.$$

The first inequality follows by counting vertices carrying two factors, one factor or no factor; it remains valid even when $f-2s$ is negative. For $t\le f = O(\log n)$, the error in Lemma 4.4 is uniformly $o(1)$, and its factor $\Gamma\le 1$ may be discarded. Using $\binom{f}{s}^2 s! \le f^{2s}/s!$, the full sum gives

$$\frac{\operatorname{per} A_T}{n!/2^n} \le (1+o(1))e^{-1}D_N(S_M)c_n^f\exp(O(f^2/N))\exp\left(\frac{2L_n^2f^2}{c_n^2N}\right).$$

All additional exponents here are $o(1)$. More explicitly, after deleting a short core subset U, the core order is $N' = n-f-|U|\sim n$, meaning $N'/n\to 1$, while the exceptional set remains the original F. The normalized core scores are still eventually at most 0.95, their squared sum is $O(\log n)$, and the cross-edge deletion size is $t\le f = O(\log n)$. Hence the error exponent in Lemma 4.4 is

$$O\left(\frac{(\log n)^{3/2}}{\sqrt{n}} + \frac{(\log n)^2}{n}\right) = o(1),$$

The constants are independent of U, I, J. The cross-edge neighbor proportions change from those in the original graph by only $O((f+|U|)/n)$, and the total one-norm deviation of the weights remains

$O(\sqrt{n \log n})$. Thus the cross-edge sum estimates above also hold uniformly. The core Gaussian factor becomes $\exp(O(|U|/n))$ times the original one: applying the Gaussian row-and-column deletion lemma to S_M/N gives logarithmic loss $O(|U|/n)$, and changing the normalization scale from N to $N - |U|$ has the same order of cost.

In the path convolution, the proportion of size-k subsets intersecting F is at most fk/n. The general permanent bound therefore bounds their full normalized short contribution by

$$K \frac{f}{\sqrt{n}} \sum_k k w_k = o(1).$$

For subsets contained in the core, restore the falling factorial and sum the positive principal minors. For every core order $N \le n$, the rank-one determinant identity gives

$$\det\left(I_N + \frac{2}{n}(I_N + A_M)\right) \le 2e \det(I_N + S_M/N).$$

To prove this, first factor out $(1 + 1/n)^N \le e$. The remaining all-ones rank-one factor is at most $1 + N/(n+1) < 2$, because the real quadratic form of the inverse of $I + S_M/(n+1)$ is at most the squared vector norm. Finally, the skew-frequency determinant product increases monotonically as the scale increases from $1/(n+1)$ to $1/N$.

Restoring the falling factorial for short subsets contributes $\exp(O(k_n^2/n)) = 1 + o(1)$. Since eventually $c_n < 1/4$, the uniform estimates above give

$$\frac{H(T)}{\mu_n} \le (1/4)^f \rho_N(S_M) + o(1) \le 3/4 + o(1) < 1, \qquad f \ge 1.$$

Since $P(n) \ge \mu_n$, a tournament with exceptional vertices cannot maximize the path count once n is sufficiently large. A weaker bound for short convolution terms avoiding F is $(10/3)4^{-f}(1 + \varepsilon_n)$. Denote the combined contribution of the other short terms and the long terms by δ_n. Then

$$\frac{H(T)}{\mu_n} \le \frac{10}{3} 4^{-f}(1 + \varepsilon_n) + \delta_n, \qquad \varepsilon_n, \delta_n \longrightarrow 0. \tag{7.1}$$

Both errors tend to zero uniformly over the low-variance class. Take n large enough that $1 + \varepsilon_n \le 21/20$ and that the short terms meeting F and the long terms each contribute at most $1/25$. For $f \ge 1$, equation (7.1) gives

$$\frac{10}{3} \cdot \frac{1}{4} \cdot \frac{21}{20} + \frac{1}{25} + \frac{1}{25} = \frac{191}{200} < 1.$$

Either choice leaves the final upper-bound constant unchanged.

An exceptional vertex satisfies

$$|d_i^+ - (n-1)/2| > \frac{9}{20}(n-1), \quad f\frac{81}{400}(n-1)^2 \le V_{\deg}(T) < 16n^2 \log n,$$

Hence $f < (6400/81)(n/(n-1))^2 \log n$. For the core,

$$\max_i |s_i^M| \le 0.9(n-1) + f, \quad \frac{0.9(n-1) + f}{n - f - 1} \longrightarrow 0.9,$$

$$\tau_M \le \frac{2\|s\|_2^2 + 2f^2 N}{(N-1)^2} = O(\log n) + O((\log n)^2/n) = O(\log n).$$

Since $|\ell_i - 1| \le |a_i|/(1-b)$ and $|r_i - 1| \le |a_i|/(1-b)$,

$$\sum_i (|\ell_i - 1| + |r_i - 1|) \le \frac{2}{1-b} \sum_i |a_i| \le \frac{2}{1-b} \sqrt{N \tau_M} = O(\sqrt{N \log n}).$$

This bounds the difference between the weighted and unweighted count of any neighbor subset by the total deviation above. Specifically,

$$\sum_j A_{xj} r_j \le N q_x + K \sqrt{N \log n}, \quad \sum_i A_{ix} \ell_i \le N(1 - q_x) + K \sqrt{N \log n}.$$

For $q \in [0, 1/20] \cup [19/20, 1]$, we have $4q(1-q) \le 4(1/20)(19/20) = 19/100$. The $O(f/n)$ perturbation of the neighbor proportion and the perturbation $\beta = o(1)$ together give $u_x v_x \le 19/100 + o(1)$.

In the four-block expansion, $t = f - s$. There are s edges within the exceptional set, t from the exceptional set to the core, t from the core to the exceptional set, and $N - t$ within the core. Their total is

$$s + t + t + (N - t) = N + f = n.$$

There are no signs in a permanent expansion, and each permutation occurs once after classification. The full powers-of-two and factorial factor is

$$\frac{[(N-t)!/2^{N-t}]\,[N^{2t}/2^{2t}]}{(N+f)!/2^{N+f}} = 2^{f-t} N^{2t} \frac{(N-t)!}{(N+f)!} = 2^s N^{2t} \frac{(N-t)!}{(N+f)!}.$$

Since

$$\frac{(N-t)!}{(N+f)!} = \prod_{h=-t+1}^{f} \frac{1}{N+h} = N^{-(f+t)} \exp\left(- \sum_{h=-t+1}^{f} \log(1 + h/N) \right),$$

and $\sum |h|/N = O(f^2/N)$, this becomes

$$2^s N^{2t-(f+t)} e^{O(f^2/N)} = (2/N)^s e^{O(f^2/N)}.$$

If $h = |R \cap C|$, the vertices carrying both u, v number $f - 2s + h$, and those carrying just one factor number $2s - 2h$. Consequently,

$$c_n^{f-2s+h} L_n^{2s-2h} = c_n^{f-2s} L_n^{2s} (c_n / L_n^2)^h \le c_n^{f-2s} L_n^{2s}.$$

Summing over all choices of internal edges gives

$$\begin{aligned} \sum_{s=0}^{f} \binom{f}{s}^2 s! (2/N)^s c_n^{f-2s} L_n^{2s} &\le c_n^f \sum_{s=0}^{f} \frac{1}{s!} \left(\frac{2 L_n^2 f^2}{c_n^2 N} \right)^s \\ &\le c_n^f \exp\left(\frac{2 L_n^2 f^2}{c_n^2 N} \right). \end{aligned}$$

Below, $\#E$ and $|E|$ both denote the cardinality of a finite set E. When $k = 0$, the deleted set avoids the exceptional set, and the proportion is zero. For $1 \le k \le n$, the proportion satisfies

$$\frac{\#\{U : |U| = k, U \cap F \ne \varnothing\}}{\binom{n}{k}} \le \sum_{x \in F} \frac{\binom{n-1}{k-1}}{\binom{n}{k}} = fk/n.$$

After multiplication by the general convolution weight $K\sqrt{n}\, w_k$, the total is $Kf \sum_k k w_k / \sqrt{n} = o(1)$.

Set $Q = S_M/(n+1)$ and $v = \mathbf{1}_N$. Then

$$\det\left(I_N+\frac{2}{n}(I_N+A_M)\right)=(1+1/n)^N\det(I_N+Q)\left[1+\frac{1}{n+1}v^{\mathsf{T}}(I_N+Q)^{-1}v\right].$$

Using $(I+Q)^{-1}=(I-Q)(I-Q^2)^{-1}$ and the vanishing of a skew-symmetric quadratic form,

$$v^{\mathsf{T}}(I+Q)^{-1}v=v^{\mathsf{T}}(I+Q^{\mathsf{T}}Q)^{-1}v\le\|v\|_2^2=N.$$

Also, $(1+1/n)^N\le e^{N/n}\le e$, and

$$\det(I+S_M/(n+1))=\prod_j\left(1+\frac{\lambda_j^2}{(n+1)^2}\right)\le\prod_j\left(1+\frac{\lambda_j^2}{N^2}\right).$$

The three factors are at most e, $\det(I+S_M/N)$, and 2, respectively. Multiplication by the permanent factor $e^{-1}/2$ leaves $\rho_N(S_M)$. The constants in the exceptional-set estimate satisfy

$$\frac{10}{3}\frac{1}{4}\frac{21}{20}=\frac{7}{8}=\frac{175}{200},\qquad\frac{1}{25}+\frac{1}{25}=\frac{16}{200},\qquad\frac{175+16}{200}=\frac{191}{200}.$$

Suppose that, after deleting the exceptional vertices and a short set, the distance of every exceptional vertex's core-neighbor proportion from $[0,1/20]\cup[19/20,1]$ is at most ϵ_n, where $\epsilon_n=O((f+k_n)/n)$. On $q\in[0,1]$, the derivative of $q(1-q)$ has absolute value at most one. Thus

$$q_x(1-q_x)\le\frac{19}{400}+\epsilon_n.$$

Using the same nonnegative weight deviation $\beta_n=O(\sqrt{\log n/n})$,

$$\begin{aligned}u_xv_x&=4(q_x+\beta_n)(1-q_x+\beta_n)\\&=4q_x(1-q_x)+4\beta_n+4\beta_n^2\\&\le\frac{19}{100}+4\epsilon_n+4\beta_n+4\beta_n^2=:c_n.\end{aligned}$$

Moreover, $u_x,v_x\le2+2\beta_n=:L_n$. Since $\epsilon_n,\beta_n\to0$, choose a uniform threshold such that

$$4\epsilon_n+4\beta_n+4\beta_n^2<\frac{6}{100},\qquad c_n<\frac{1}{4}.$$

Each term in the additional exponent above also tends to zero:

$$\frac{f^2}{N}=O((\log n)^2/n)\to0,\quad\frac{2L_n^2f^2}{c_n^2N}=O((\log n)^2/n)\to0.$$

For $\tau\le A\log n,t\le B\log n$, the error terms in Lemma 4.4 are bounded respectively by constant multiples of

$$\frac{\log n+(\log n)^2}{n},\quad\frac{(\log n)^{3/2}}{\sqrt{n}},\quad\sqrt{\frac{\log n}{n}},\quad\frac{\log n+1}{n},\quad\frac{(\log n)^{3/2}}{\sqrt{n}},\quad\frac{(\log n)^2}{n}.$$

The constants are independent of the tournament and deletion sets, so the resulting $o(1)$ error is uniform.

7.4 The score penalty in the remaining class

Proposition 7.3 (Uniform score-sensitive bound). Fix $K_0 > 0$. There are constants K, N depending only on K_0 such that, for $n \geq N$, every tournament with $a = S\mathbf{1}/(n-1)$, $\|a\|_\infty \leq 0.9$, and $\tau = \sum_i a_i^2 \leq K_0 \log n$ satisfies

$$\frac{H(T)}{\mu_n} \leq \rho_n(S) \exp\left(-\frac{\tau}{2} + \frac{K}{n}\right) + B_n,$$

where $B_n \geq 0$ is the graph-independent tail from Lemma 5.1a and $B_n = o(n^{-1})$.

Proof.

The remaining tournaments satisfy

$$\max_i |a_i| \leq 0.9, \qquad a = s/(n-1), \qquad \tau = \sum_i a_i^2 \leq K \log n.$$

For the fixed K_0 in the statement, all subsequent constants may depend on K_0. The paired product satisfies

$$\Gamma = \prod_i (1 - a_i^2) \leq e^{-\tau}.$$

Apply Lemma 4.4 to the whole tournament. The score product is common to every short term. Retaining it outside the sum absorbs errors that, individually, may exceed $1/n$.

For a principal deleted set of size $k \leq k_n$, denoted by U, Lemma 4.4 gives

$$\operatorname{per} A[U^c] \leq e^{-1} D_n(S) \Gamma \left(\prod_{i \in U} g_i\right) \frac{(n-k)!}{2^{n-k}} \exp(R_\tau + R_{\tau,k}),$$

$$g_i = (1 - a_i^2)^{-1} \leq G := 100/19,$$

$$R_\tau \leq K[\mathcal{M}_\tau + \sqrt{\tau/n} + 1/n], \qquad R_{\tau,k} \leq K[k\sqrt{\tau/n} + (k + k^2)/n].$$

When accumulating errors, we must retain the common factor Γ. Define the nonnegative coefficients

$$c_k = (2/n)^k \sum_{|U|=k} \det(I + A[U]) \prod_{i \in U} g_i.$$

They satisfy $c_0 = 1$ and $c_k \leq G^k w_k$. Restoring the falling factorial produces

$$\log \frac{n^k}{(n)_k} = O(k^2/n), \qquad k \leq k_n.$$

Set $\delta_n = \sqrt{\tau/n} + 1/n$ and $r_k = R_{\tau,k} + \log(n^k/(n)_k)$. Taking a nonnegative error bound, uniformly for $\tau \leq K_0 \log n$ we have $0 \leq r_k \leq K\delta_n(k + k^2)$ and $\max_{k \leq k_n} r_k = o(1)$. Therefore

$$\sum_{k \leq k_n} c_k e^{r_k} \leq \sum_{k=0}^{n} c_k + K\delta_n \sum_{k \geq 0} G^k w_k (k + k^2) \leq \left(\sum_{k=0}^{n} c_k\right) e^{K\delta_n}.$$

The last step uses $\sum_k c_k \geq 1$ and convergence of the corresponding fixed moments. The normalized short path sum is consequently at most

$$\frac{e^{-1} D_n(S) \Gamma}{2} e^{R_\tau + K\delta_n} \det\left(I + \frac{2}{n} \operatorname{diag}(g)(I + A)\right).$$

The total error is controlled by the finite moments of $G^k w_k$, rather than by the maximum deleted-set size k_n. This avoids an extra logarithmic factor in the final error.

We now remove the weights while preserving the score penalty. Set $W = I + (2/n)(I + A)$ and $\Delta = (2/n)\,\mathrm{diag}(g-1)(I+A)$. The symmetric part of W is at least I, so $\|W^{-1}\|_{\mathrm{op}} \le 1$. Each row of $I + A$ has Euclidean norm at most $\sqrt{n}$, and $\sum_i (g_i - 1) \le K\tau$. Decomposing Δ as a sum of rank-one row matrices therefore controls its trace norm, the sum of its singular values. Using $\|uv^{\mathsf{T}}\|_* = \|u\|_2\|v\|_2$, we obtain

$$\|\Delta\|_* \le K\tau/\sqrt{n}.$$

The determinant inequality $|\det(I+E)| \le \exp(\|E\|_*)$ gives

$$\det(W+\Delta) \le \det(W) e^{K\tau/\sqrt{n}}.$$

Both determinants are positive: the symmetric part of W is positive definite, and the weighted determinant has a principal-minor expansion with positive coefficients. The unweighted rank-one bound of Section 7.3, with $N = n$, is $\det(W) \le 2e\det(I + S/n)$.

Combining the factors above and adding only the long-deletion tail at the end yields the uniform score-sensitive upper bound

$$\frac{H(T)}{\mu_n} \le \rho_n(S) \exp\left\{ -\tau + K\left[\frac{1+\tau+\tau^2}{n} + \frac{\tau^{3/2}+\tau}{\sqrt{n}} + \sqrt{\tau/n} \right] \right\} + B_n. \tag{7.2}$$

For $\tau \le K_0 \log n$ and sufficiently large order, the polynomial error terms other than the constant $1/n$ consume at most $\tau/4$ in total. Young's inequality controls the remaining square-root term:

$$K\sqrt{\tau/n} \le \tau/4 + K^2/n.$$

The total exponent in equation (7.2) is therefore at most $-\tau/2 + K/n$. Applying Proposition 6.6 gives

$$H(T)/\mu_n \le r_n e^{K/n} + B_n = L + O(1/n).$$

This proves Proposition 7.3 and the sharp bound in the remaining class. □

7.5 Completion of the upper bound

The three classes cover all tournaments. High-variance tournaments have normalized path count $o(1/n)$; low-variance tournaments containing exceptional vertices have count strictly below one; and every remaining tournament satisfies the bound $L + O(1/n)$ just proved. Hence

$$P(n) \le \big(L + O(1/n)\big)\mu_n.$$

The threshold in each case is independent of the particular tournament. Taking a common threshold proves the upper bound in Proposition 7.1.

8. The small-score formula and the lower bound

Theorem 8.1 (Small-score path approximation).

There is an absolute constant C with the following property. For every nonnegative function satisfying $d(n)/\sqrt{n} \to 0$, denoted by $d = d(n)$, there is a threshold $n_0(d)$ such that every tournament of order $n \ge n_0(d)$ with $\|S\mathbf{1}\|_\infty \le d(n)$ satisfies

$$\left| \frac{H(T)}{\mu_n} - \rho_n(S) \right| \le C\frac{(d(n)+1)^2}{n}.$$

8.1 Proof of the small-score approximation

Proof.

Write $d = d(n)$. Apply Lemma 4.5 to every principal deleted set $|U| = k \le k_n$. Since $d + k_n + 1 = o(\sqrt{n})$, it applies uniformly over all these subsets. After restoring the falling factorial $(n)_k$, the positive path convolution gives

$$\frac{H(T)}{\mu_n} = \frac{e^{-1}D_n(S)}{2} \sum_{k \le k_n} \left(\frac{2}{n}\right)^k \sum_{|U|=k} \det(I + A[U]) + O\left(\frac{(d+1)^2}{n}\right).$$

This error rate has no extra logarithmic factor, because the error is at most a fixed constant times

$$\frac{1}{n} \sum_{k \ge 0} w_k (d + k + 1)^2 = O\left(\frac{(d+1)^2}{n}\right).$$

Restoring the falling factorial contributes $O(n^{-1} \sum k^2 w_k)$, and Section 5.1 bounds the terms with large deleted sets by $o(n^{-1})$. Extending the short generating sum above to all k costs $n^{-2+o(1)}$, because its coefficients are bounded by w_k. Thus

$$\frac{H(T)}{\mu_n} = \frac{e^{-1}D_n(S)}{2} \det\left(I + \frac{2}{n}(I + A)\right) + O\left(\frac{(d+1)^2}{n}\right).$$

Set $u = \mathbf{1}/\sqrt{n}$ and $Q = S/(n+1)$. The rank-one determinant lemma gives

$$\det\left(I + \frac{2}{n}(I + A)\right) = (1 + 1/n)^n \det(I + Q) \left(1 + \frac{n}{n+1} u^{\mathsf{T}} (I + Q)^{-1} u\right).$$

Skew-symmetry gives

$$u^{\mathsf{T}}(I + Q)^{-1}u = u^{\mathsf{T}}(I - Q^2)^{-1}u,$$

The absolute difference of this quantity from one is at most

$$\|Qu\|_2^2 \le \frac{d^2}{(n+1)^2}.$$

Also, $(1+1/n)^n = e(1+O(n^{-1}))$. Replacing $S/(n+1)$ by S/n incurs a logarithmic determinant error $O(n^{-1})$, by the bounded sum of the squared normalized frequencies. Consequently,

$$\det\left(I + \frac{2}{n}(I + A)\right) = 2e \det(I + S/n) \left(1 + O(n^{-1} + d^2/n^2)\right).$$

Substitution proves the theorem. All estimates use the same absolute constant; only the threshold beyond which the small-score condition holds depends on d.

For every k, using $d + k + 1 \le (d+1)(k+1)$ gives

$$\frac{1}{n} \sum_k w_k (d + k + 1)^2 \le \frac{(d+1)^2}{n} \sum_k w_k (k+1)^2.$$

Thus using k_n as a replacement for k is unnecessary, and no $\log n$ loss occurs. For $Q^{\mathsf{T}} = -Q$,

$$(I + Q)^{-1} = (I - Q)(I - Q^2)^{-1}, \qquad Q(I - Q^2)^{-1} \text{ is skew-symmetric},$$

$$\begin{aligned}1 - u^{\mathsf{T}}(I+Q)^{-1}u &= u^{\mathsf{T}}[I-(I+Q^{\mathsf{T}}Q)^{-1}]u \\ &= (Qu)^{\mathsf{T}}(I+QQ^{\mathsf{T}})^{-1}(Qu) \le \|Qu\|_2^2.\end{aligned}$$

Since $Qu = s/[(n+1)\sqrt{n}]$,

$$\|Qu\|_2^2 = \frac{\sum_i s_i^2}{n(n+1)^2} \le \frac{d^2}{(n+1)^2}.$$

The determinant error from changing the normalization satisfies

$$\begin{aligned}0 &\le \log\det(I+S/n) - \log\det(I+S/(n+1)) \\ &= \sum_j \int_{\lambda_j^2/(n+1)^2}^{\lambda_j^2/n^2} \frac{du}{1+u} \\ &\le \sum_j \lambda_j^2 \left(\frac{1}{n^2} - \frac{1}{(n+1)^2}\right) \\ &= \frac{n(n-1)}{2}\frac{2n+1}{n^2(n+1)^2} = O(n^{-1}).\end{aligned}$$

and

$$0 \le 1 - n\log(1+1/n) = n\int_0^{1/n} \frac{u}{1+u}\,du \le \frac{1}{2n}.$$

The difference between the rank-one factor and two is

$$\left|1 + \frac{n}{n+1}u^{\mathsf{T}}(I+Q)^{-1}u - 2\right| \le \frac{1}{n+1} + \frac{d^2}{(n+1)^2}.$$

Multiplying in order and using $D_n(S)\det(I+S/n) < 3$ gives the absolute error in Theorem 8.1. □

8.2 Carousel tournaments

Corollary 8.2 (Carousel lower bound).

For the odd-order carousel tournaments and even-order one-vertex-deletion constructions defined below,

$$H(\mathrm{Car}_n)/\mu_n = L + O(n^{-1}).$$

Proof.

For odd n, take the residue classes modulo n and define the carousel tournament Car_n as follows: for $1 \le j \le (n-1)/2$, direct $i \to i+j$. It is regular, so $d = 0$. For even n, let Car_n be obtained by deleting one vertex from Car_{n+1}. Then $d = 1$.

The sign matrix of an odd-order carousel is similar to T_n^0 by a signed permutation. Explicitly, use the diagonal signs $(-1)^i$, $0 \le i < n$, to conjugate T_n^0, and reorder the indices as $0, 2, \ldots, n-1, 1, 3, \ldots, n-2$. When two indices have the same parity, the switched edge follows increasing index order; when they have different parities, its direction is reversed. Reading indices in the stated cyclic order gives the carousel orientation. Restricting the inverse signed conjugation and reordering after a one-vertex deletion gives T_n^0 on the remaining ordered indices. Thus the same signed spectral calculation remains valid after any one vertex is deleted, and both parity cases satisfy

$$\rho_n(S_{\mathrm{Car}_n}) = r_n := \frac{(n+1)^n + (n-1)^n}{(n+i)^n + (n-i)^n}.$$

Divide numerator and denominator by n^n. At each fixed $z = 1, -1, i, -i$, use the Taylor expansion of $(1 + z/n)^n$. This gives

$$r_n = \frac{\cosh 1}{\cos 1} + O(n^{-1}) = L + O(n^{-1}).$$

Both parities therefore have $H(\mathrm{Car}_n)/\mu_n = L + O(n^{-1})$, and hence

$$P(n) \geq (L - O(n^{-1}))\mu_n.$$

The transition from signed similarity to the path count uses Theorem 8.1, with $d = 0$ and $d = 1$, respectively. Let the constants and thresholds for the lower and upper bounds be K_l, N_l and K_u, N_u. Taking $K = \max(K_l, K_u)$ and $n_0 = \max(N_l, N_u, 2)$ completes Theorem 1.1.

Write $n = 2m + 1$, with original indices $i = 0, \ldots, 2m$. Let $D_{ii} = (-1)^i$. For $i < j$,

$$(DT_n^0 D)_{ij} = (-1)^{i+j}.$$

Use the reordering $p_k = 2k \pmod n$, represented in $0, \ldots, n-1$. It gives the sequence $0, 2, \ldots, 2m, 1, 3, \ldots, 2m-1$. If i, j have the same parity and their positive difference is $2h$, then

$$2^{-1}(j - i) \pmod n = h \in \{1, \ldots, m\}.$$

If their parities differ and $i < j$, write $j - i = 2h + 1$. Then

$$2^{-1}(j - i) \pmod n = h + m + 1 \in \{m + 1, \ldots, 2m\}.$$

In the new indices, edges with differences $1, \ldots, m$ point forward, and those with differences $m + 1, \ldots, 2m$ point backward, exactly as in the carousel tournament. A one-vertex deletion selects the corresponding principal submatrix. The restricted D is still a diagonal sign matrix, and the transitive principal submatrix on the remaining ordered indices is still T_n^0. This identifies the spectrum of both parity constructions. Their path counts are then estimated separately by Theorem 8.1.

By $D_n(S) = \det(I + iS/n)^{-1}$ and the principal-minor polynomial,

$$\begin{aligned}\rho_n(S) &= \frac{\det(I + S/n)}{\det(I + iS/n)} \\ &= \frac{(1 + 1/n)^n + (1 - 1/n)^n}{(1 + i/n)^n + (1 - i/n)^n} = r_n.\end{aligned}$$

For $z \in \{1, -1, i, -i\}$ and $n \geq 2$, the power series for the principal logarithm on $|w| < 1$ gives

$$\begin{aligned}n \log(1 + z/n) &= z - \frac{z^2}{2n} + R_{n,z}, \\ |R_{n,z}| &\leq \sum_{h=3}^{\infty} \frac{1}{hn^{h-1}} \leq \frac{1}{3n^2(1 - 1/n)} \leq \frac{2}{3n^2}.\end{aligned}$$

Exponentiating yields $(1 + z/n)^n = e^z(1 - z^2/(2n) + O(n^{-2}))$. Adding the terms in pairs gives

$$\begin{aligned}(1 + 1/n)^n + (1 - 1/n)^n &= 2\cosh 1\,[1 - 1/(2n) + O(n^{-2})], \\ (1 + i/n)^n + (1 - i/n)^n &= 2\cos 1\,[1 + 1/(2n) + O(n^{-2})].\end{aligned}$$

Because $\cos 1 > 0$, the denominator is bounded below by a fixed positive constant for all sufficiently large n. Using

$$\frac{1}{1+u} = 1 - u + \frac{u^2}{1+u},$$

we obtain the more precise spectral ratio

$$r_n = L[1 - 1/n + O(n^{-2})].$$

The term $-L/n$ is the first correction to the spectral ratio. Identifying the corresponding coefficient for the path count requires an error smaller than the $O(n^{-1})$ estimate in Theorem 8.1. Finally,

$$H(\mathrm{Car}_n)/\mu_n = r_n + O(n^{-1}) = L + O(n^{-1}),$$

there is therefore a fixed K_l such that $P(n) \geq (L - K_l/n)\mu_n$. □

8.3 Degree stability near the maximum

Corollary 8.2a (Degree-variance stability). There are absolute constants C, N such that, for $n \geq N$ and $0 \leq \varepsilon \leq (L-1)/2$, any tournament satisfying $H(T)/\mu_n \geq L - \varepsilon$ has

$$\tau(T) \leq C(\varepsilon + n^{-1}), \qquad V_{\deg}(T) \leq C(\varepsilon n^2 + n).$$

In particular, every tournament maximizing $H(T)$ has $V_{\deg}(T) = O(n)$.

Proof. The assumption gives $H(T)/\mu_n \geq (L+1)/2 > 1$. The first two classes of Section 7 are therefore excluded by a uniform threshold. Proposition 7.3 and Proposition 6.6 give

$$L - \varepsilon - B_n \leq (L + K_r/n)\exp(-\tau/2 + K/n).$$

For sufficiently large n, $B_n \leq 1/n$ and the left-hand side is bounded below by $(L+1)/4$. Taking logarithms and using $\log(1+x) \leq x$ proves $\tau \leq C(\varepsilon + 1/n)$. Multiply by $(n-1)^2/4$ for the variance estimate. After enlarging the threshold so that $K_l/n \leq (L-1)/2$, Corollary 8.2 permits $\varepsilon = K_l/n$ for an exact maximizer. □

The conclusion controls mean square degree deviation and gives a maximum single-vertex deviation of order $\sqrt{n}$. Eventual regularity, carousel uniqueness, edit-distance stability, and the $o(\sqrt{n})$ condition in Theorem 8.1 each require further arguments.

9. The Hessian of the original angular function

Let A be the adjacency matrix of a regular tournament of odd order $d \geq 3$, and put

$$S = 2A + I - J, \quad C = S/d, \quad P_0 = J/d, \quad z = 2/d.$$

Thus $C^T = -C$ and $C\mathbf{1} = 0$. Define

$$B(y) = I - (A + \tfrac{1}{2}I)\,\mathrm{diag}(y), \qquad K(y) = \det B(y).$$

Near $z\mathbf{1}$, eliminate y_d from $K(y) = 0$, writing $y_d = h(y_1, \ldots, y_{d-1})$. The angular function uses the local logarithm with value $\log z$ at the base point:

$$\widetilde{f}(\theta) = \log h(ze^{i\theta_1}, \ldots, ze^{i\theta_{d-1}}).$$

Theorem 9.2 (Original angular Hessian). The elimination and logarithm are analytic near the base point. With $E = (e_1 - e_d, \ldots, e_{d-1} - e_d)$,

$$\mathbf{H} := \nabla^2 \widetilde{f}(0) = E^T Q E \succ 0, \qquad Q = (I + C^2)(I - C^2)^{-1}.$$

In particular, writing $\mathcal{H} = \det \mathbf{H}$,

$$\mathcal{H} = d\frac{\det(I + C^2)}{\det(I - C^2)}, \qquad \sqrt{d/\mathcal{H}} = \frac{\det(I + C)}{\det(I + iC)} = \rho_d(S) \in \mathbb{Q}_{>0}.$$

Proof. Let $u = \mathbf{1}/\sqrt{d}$, and let the columns of V be an orthonormal basis of $u^\perp$. Every nonzero frequency λ of C satisfies

$$\lambda^2 \le \tfrac{1}{2}\|C\|_F^2 = \frac{d-1}{2d} < \frac{1}{2}.$$

Set $P = (2A + I)/d = P_0 + C$ and introduce $y_j = ze^{t_j}$. In the orthonormal basis (u, V) write

$$(u, V)^T [I - P\,\mathrm{diag}(e^t)](u, V) = \begin{pmatrix} a & b \\ c & D \end{pmatrix}.$$

At zero, $a = b = c = 0$ and $D(0) = V^T(I - C)V$ is invertible. Thus the analytic Schur complement

$$\psi(t) = a(t) - b(t)D(t)^{-1}c(t)$$

has the same local zero set as $K(ze^t)$. Put $U = (I - C)^{-1}$. Differentiation at zero gives

$$a_j = -1/d, \quad a_{jk} = -\delta_{jk}/d, \quad b_j = -e_j^T V/\sqrt{d}, \quad c_k = -V^T C e_k/\sqrt{d}.$$

Since $b = c = 0$, only $b_j D^{-1} c_k + b_k D^{-1} c_j$ survives from the second derivative of $bD^{-1}c$. Using $VD(0)^{-1}V^T C = UC$, we obtain

$$\psi_j(0) = -1/d, \qquad \psi_{jk}(0) = -\frac{\delta_{jk} + (UC)_{jk} + (UC)_{kj}}{d} = -Q_{jk}/d,$$

where $I + UC + (UC)^T = (I + C^2)(I - C^2)^{-1}$. Moreover, with $q = \det(I - C) > 0$,

$$K_{y_d}(z\mathbf{1}) = -q/(dz) = -q/2 \neq 0.$$

The analytic implicit function theorem therefore supplies h, with $h(z, \dots, z) = z$. Choose the logarithm locally and set

$$g(t') = \log[h(ze^{t'})/z], \qquad \psi(t', g(t')) = 0.$$

Implicit differentiation yields $g_j(0) = -1$ and

$$g_{jk}(0) = -(Q_{jk} - Q_{jd} - Q_{dk} + Q_{dd}).$$

The substitution $t' = i\theta$ gives $\widetilde{f}(\theta) = \log z + g(i\theta)$ and $\nabla^2 \widetilde{f}(0) = -\nabla^2 g(0) = E^T Q E$, with the sign supplied by $i^2 = -1$.

On each skew block of frequency λ, Q acts by $(1-\lambda^2)/(1+\lambda^2) > 0$, and $Qu = u$. Therefore $Q \succ 0$ and $E^T Q E \succ 0$. The columns of E are not orthonormal: $E^T E = I_{d-1} + J_{d-1}$ has determinant d. Consequently

$$\det(E^T Q E) = d \det_{u^\perp} Q = d \det Q.$$

Finally, $\det(I+C^2)=\det(I+iC)^2$ and $\det(I-C^2)=\det(I+C)^2$. Both unsquared determinants are positive, as follows blockwise from $1-\lambda^2>0$ and $1+\lambda^2>0$. Odd principal minors of the rational skew matrix C vanish; hence $\det(I+iC)$ is rational, proving the last assertion. □

The calculation identifies the full Hessian in the original angular coordinates. The expanded Schur-complement calculations and the explicit local elimination are retained in Detailed proof companion §9.

10. Model limits and the carousel constants

10.1 Four fixed-base limits

Fix a regular base tournament of odd order $d \geq 3$. Replace its vertices by independent uniform labeled tournaments of order m (model $\mathcal{T}$), or by independent uniform labeled regular tournaments of odd order m (model $\mathcal{R}$). Edges between blocks follow the base. With $N=dm$, normalize cycle and path expectations by

$$\nu_N=(N-1)!/2^N, \qquad \mu_N=N!/2^{N-1}.$$

Corollary 10.2. For fixed d, the two limits for model $\mathcal{T}$ are $\rho_d(S)$, and those for model $\mathcal{R}$ are $e^{1/d}\rho_d(S)$:

$$\frac{\mathbb{E}C_{\mathcal{T}}}{\nu_N},\ \frac{\mathbb{E}H_{\mathcal{T}}}{\mu_N}\longrightarrow\rho_d(S), \qquad \frac{\mathbb{E}C_{\mathcal{R}}}{\nu_N},\ \frac{\mathbb{E}H_{\mathcal{R}}}{\mu_N}\longrightarrow e^{1/d}\rho_d(S).$$

The regular-block limits are taken through odd m.

Proof. Wormald's Theorem 4 gives the first cycle limit as $\sqrt{d/\mathcal{H}}$ when $\mathcal{H}\neq 0$, now established by Theorem 9.2. His Theorem 5 transfers it to paths, and Theorems 1 and 5 supply the regular-block factor $e^{1/d}$. □

For comparison, Divoux–Wang [12, Proposition 5.1] gives the same constants directly. For $W=(J_N-I_N+S\otimes J_m)/2$, put $D=2W/(N-1)$ and $B=D-J_N/N$. On the constant direction, the block-constant zero-sum space, and the within-block zero-sum space, B acts respectively as

$$0, \qquad B_0=\left.\frac{mS-I_d}{N-1}\right|_{\mathbf{1}_d^{\perp}}, \qquad -I/(N-1).$$

Consequently its operator norm stays below one, and the determinant ratio in their counting formula satisfies

$$R_N=\frac{\det(I-B)}{\sqrt{\det(I-B^TB)}}\longrightarrow e\rho_d(S).$$

Rescaling D to W contributes $((N-1)/N)^N\to e^{-1}$. Their auxiliary-vertex construction gives $\mathrm{path}(D)=(1+o(1))N^2\,\mathrm{cyc}(D)$, and Wormald's comparison supplies $e^{1/d}$ for regular blocks. Detailed proof companion §10.1 records the determinant factors and the auxiliary-vertex normalization. Theorem 9.2 additionally identifies the original angular Hessian itself.

10.2 Closed form and monotonicity

Proposition 10.3. For odd $d\geq 3$, put $r_d=\rho_d(\mathrm{Car}_d)$ and $\mathfrak{r}_d=e^{1/d}r_d$. Then

$$r_d=\frac{(1+1/d)^d+(1-1/d)^d}{(1+i/d)^d+(1-i/d)^d}.$$

Indeed, the carousel sign matrix is signed-permutation similar to the transitive sign matrix T_d. Its determinant polynomial is

$$\det(I+tT_d)=\frac{(1+t)^d+(1-t)^d}{2};$$

taking $t = 1/d$ and i/d proves the formula. These determinant evaluations are from [12, Section 7 and Lemma 9.1]; the elementary calculations are recorded in Detailed proof companion Appendix A.1–A.2.

Corollary 10.5. The sequence $\mathfrak{r}_d$, over odd $d \geq 3$, is strictly increasing and tends to $L = \cosh 1/\cos 1$.

Proof. The positive frequencies of T_d/d are [11, Lemma 4.2]

$$a_{j,d} = d^{-1} \cot \frac{(2j-1)\pi}{2d}, \qquad 1 \leq j \leq (d-1)/2.$$

Expanding the logarithmic determinant ratio and using $2\sum_j a_{j,d}^2 = 1 - 1/d$ gives

$$\log \mathfrak{r}_d = 1 + 2 \sum_{j=1}^{(d-1)/2} \sum_{r \geq 1} \frac{a_{j,d}^{4r+2}}{2r+1}.$$

For fixed j, $a_{j,d}$ increases with d: writing $c_j = (2j-1)\pi/2$ and $x = c_j/d$, it equals $x \cot x/c_j$, and

$$(x \cot x)' = \frac{\sin x \cos x - x}{\sin^2 x} < 0 \quad (0 < x < \pi/2).$$

Passing from d to $d+2$ increases every existing frequency and adds a positive one. The displayed convergent series has positive coefficients, proving strict increase. The closed form gives the limit L. □

For completeness, the higher-order expansion is

$$\mathfrak{r}_d = L \left[1 + \frac{a_2}{d^2} + \frac{a_4}{d^4} - \frac{1}{3d^5} + O(d^{-6}) \right],$$

$$a_2 = \frac{\tanh 1 - \tan 1}{3}, \qquad a_4 = \frac{\tanh 1 + \tan 1}{5} + \frac{2 + \tan^2 1 - \tanh^2 1}{18} + \frac{a_2^2}{2}.$$

This follows by expanding the closed form at $x = 1/d = 0$; uniform Taylor remainder bounds are retained in Detailed proof companion §10.4. The order of limits matters: the block limit is first taken at fixed d, and only then does d tend to infinity.

11. The unique expected extremum for a fixed base order

The models of §10 are now defined for *every* tournament base of a fixed odd order $d \geq 3$, without assuming regularity. The normalization is unchanged.

Corollary 11.2. There is a threshold M_d such that, for every $m \geq M_d$, the carousel base uniquely maximizes each of the four expected counts (cycles and paths in $\mathcal{T}$ and $\mathcal{R}$), up to ordinary tournament isomorphism. In $\mathcal{R}$, only odd m are admissible.

The assertion concerns expected counts. The threshold depends on d, but is common to all bases of that order and all four admissible comparisons.

11.1 Nonregular bases have exponentially small normalized expectations

Let d_i be the base outdegrees. Set

$$b_i = d_i + \tfrac{1}{2}, \quad a_i = 2b_i/d = 1 + s_i/d, \quad s_i = 2d_i - (d-1), \quad \kappa = (\prod_i a_i)^{1/d}.$$

Here $0 < a_i < 2$ and $\sum_i a_i = d$. The bound $\log a \leq a - 1 - (a-1)^2/8$ on $(0, 2)$ implies

$$\log \kappa \leq -\frac{\sum_i s_i^2}{8d^3}.$$

For a nonregular base, the zero-sum even integers s_i have squared sum at least 8, so $\kappa < 1$ and $\kappa^N \le e^{-N/d^3}$.

In model $\mathcal{T}$, the expected adjacency matrix W has row sum $r_i = mb_i - 1/2$ for each row of type i. A Hamilton cycle uses distinct unordered pairs, so independence of internal edges gives $\mathbb{E}C_{\mathcal{T}} = C(W) \le \operatorname{per} W$.

For a matrix with entries in $[0,1]$, rowwise linear optimization shows that its permanent is bounded by that of a zero–one matrix whose row sums are the rounded-up original sums. Indeed, first increase a row to the target integer sum, then maximize its linear permanent functional on the corresponding hypersimplex; a maximum occurs at a zero–one vertex. Apply this successively to all rows. Brégman's inequality consequently gives

$$\operatorname{per} W \le \prod_{i=1}^{d} (R_i!)^{m/R_i}, \qquad R_i = \lceil mb_i - \tfrac{1}{2} \rceil.$$

For fixed d, $R_i = mb_i + O(1)$ and $R_i \ge m/4$ when $m \ge 2$. Stirling's bounds therefore yield

$$\log \frac{\operatorname{per} W}{N!/2^N} \le N \log \kappa + O_d(\log N).$$

Thus $\mathbb{E}C_{\mathcal{T}}/\nu_N \le N^{O_d(1)}\kappa^N \to 0$. In model $\mathcal{R}$, internal edges are not independent; instead every sample has row sum exactly $mb_i - 1/2$. Apply Brégman's inequality to each sample adjacency matrix, then average, obtaining the same bound. Explicit constants are retained in Detailed proof companion §§11.4–11.5.

To treat paths, add a vertex with independent fair edges to every old vertex. For every tournament G of order $N \ge 2$,

$$\mathbb{E}[C_{\text{aug}} \mid G] = H(G)/4, \qquad 4\nu_{N+1} = \mu_N.$$

The two closing edges of a path are distinct, which is why the factor is exactly $1/4$. In the augmented expectation matrix for $\mathcal{T}$, old rows have sum mb_i; in every augmented sample of $\mathcal{R}$, they have sum at most $mb_i + 1/2$. The common rounded targets $R_i' = \lceil mb_i + 1/2 \rceil = mb_i + O(1)$, together with target N for the new row, again allow Brégman's inequality. The old rows contribute $N \log \kappa + O_d(\log N)$ after normalization, and the one new row changes only the polynomial factor. Hence

$$\frac{\mathbb{E}H_{\mathcal{T}}}{\mu_N}, \quad \frac{\mathbb{E}H_{\mathcal{R}}}{\mu_N} \le (N+1)^{O_d(1)}\kappa^N \longrightarrow 0.$$

For a fully explicit normalization of this augmentation bound, see Detailed proof companion §11.6. The degree-based bounds above cover every nonregular base.

11.2 Strict comparison and one common threshold

For regular bases, Corollary 10.2 and the equality classification in §6 show that the carousel has strictly larger limiting constants than every nonisomorphic regular base. For regular tournaments, the equality classification yields ordinary isomorphism, which is the equivalence relevant to the blow-up models. Nonregular bases have limit zero by §11.1, whereas the carousel limits r_d and $e^{1/d}r_d$ are positive.

There are finitely many bases at fixed order d. For each noncarousel base and each of the four expectations, strict separation of the limits supplies an eventual strict inequality. Taking the maximum of this finite set of thresholds gives M_d, with odd m used for $\mathcal{R}$. Finally, a base isomorphism merely relabels the blocks and preserves their independent identical distributions and all path and cycle counts. Thus carousel-isomorphic bases give equal expectations, and only those do. This proves Corollary 11.2. □

Author statement

The paper incorporates two independently developed projects. The path framework and its L/C_* bound are in the version-specific archive [14]; the Wormald project and its overlap with [13,16] are described in the introduction. Xiangyu Ye completed the companion Lean formalization of the earlier path bound, whose precise statement is recorded in the earlier fixed repository revision.

The repository snapshot [17] documents Lean endpoints for 30 numbered results of the pre-revision manuscript, including the sharp path bound, original angular Hessian, path convolution, spectral comparison and fixed-base expectation extrema. Its coverage table specifies objects, hypotheses, quantifiers and coverage boundaries. The associated reproduction instructions fix the toolchain and audit procedure.

This revision changes the exposition and adds explicitly identified corollaries. The Lean coverage and source hashes refer to the historical snapshots above. Logs and scope notes for the present rerun of the fixed project accompany this revision; a new endpoint mapping for the concise manuscript remains to be completed. AI assistance was used in exposition and parts of proof checking. The blow-up extremality threshold is existential for each fixed odd base order; neither a computable threshold nor uniformity in growing base order is asserted.

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